

The Allocation of Teaching Talent and Human Capital Accumulation

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Abstract

The educational landscape in the U.S. has gone through major changes since the end of World War II. Real expenditures per student have risen from approximately \$2,100 to more than \$10,000 by the turn of the century and more than \$12,000 since the Great Recession. At the same time, the student-teacher ratio has fallen from a national average of almost 27 in 1955 to 16 by the 2010s. Yet despite the rise in expenditures and the reduction in class sizes, educational outcomes in the U.S. don't compare very favorably with countries at similar income levels. One aspect of U.S. education that has not garnered a lot of attention until fairly recently is occupational choice. We add an education sector to an otherwise standard ?-style model to explore the extent to which changes in career opportunities in other occupations affect the selection of workers into teaching careers. In our model, changes in the allocation of teaching talent have implications for the evolution of class size as well as quality of instruction and hence the accumulation of human capital during the workers' formative years. This gives rise to a trade-off between static and dynamic efficiency, which we quantify by way of a structural model. In order to discipline the parameterization of the model, we compare model-generated moments to their empirical counterparts in three longitudinal surveys: Project TALENT, the NLSY79, and NLSY97. Project TALENT surveys 377,000 U.S. high school students in 1960 and follows them for more than a decade. Together with the two NLSY's, it provides a detailed account of, among many others, the educational trajectories and career paths of young American adults over the course of more than four decades.

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1 Introduction

The educational landscape in the U.S. has gone through major changes since the end of World War II. Real expenditure per student has risen from approximately \$2,100 to more than \$10,000 by the turn of the century and more than \$12,000 since the Great Recession. At the same time, the student-teacher ratio has fallen from a national average of almost 27 in 1955 to 16 by the 2010s. Yet despite the rise in expenditures and the reduction in class sizes, educational outcomes in the U.S. don't compare very favorably with countries at similar income levels.

In this paper, we develop a novel unified theory of occupational choice and human capital formation with the aim of reconciling several salient stylized facts. In addition to the change in the student-teacher ratios and per-student-expenditures mentioned above, they are: (1) the increase in the share of women in teaching jobs from 4.6 percent in 1970 to 6.7 percent in 2010; (2) the drop in the share of men who are teachers from 2.9 percent to 2.1 percent during that same time; (3) the sharp rise in the female labor force participation rate; (4) the slight decline in the male labor force participation rate; and (5) the evolution of the skill composition by gender and occupation between 1970 and 2010.¹

In order to account for these stylized facts, we add an education sector to an otherwise standard β -style model to explore the extent to which changes in career opportunities in other occupations – including home production – affect the selection of workers into teaching careers. In our model, teaching is distinct from other occupations in the economy. The teacher's human capital is one of several inputs into an education technology through which students of different abilities accumulate human capital. Human capital, in turn, is the key input for the production of final goods and services. In other words, workers in non-teaching occupations produce final output while teachers contribute to the production of human capital in today's students who, in turn, will either produce units of final output next period or train the next generation of students.

In contrast to versions of a β -type model that have been used widely in the previous literature, we allow for the possibility that occupational choice is shaped by comparative as well as absolute advantage.² The elasticity of income with respect to the worker's human capital can be different from unity and this implies that two workers with identical *comparative* but different *absolute* advantages (i.e., they can be ranked "vertically"), may select distinct occupations. This variation gives us additional flexibility to account for the evolution of em-

¹The secular rise in educational achievement as well as gender- and occupation-specific variations in this trend are additional salient aspects that merit further attention. As will become apparent in sections 2 and 3, the current version of our model is not particularly well-equipped to address this stylized fact. This is an area for future work.

²Roy's original contribution accounted for selection based on absolute and comparative advantage. Some later work emphasized comparative over of absolute advantage.

ployment shares across occupations separately from changes in the distribution of skills or human capital across jobs.

The question of whether low or high-skill workers are more like to enter (or exit, for that matter) the teaching profession is critical for understanding whether static efficiency gains associated with the elimination of labor market discrimination and educational barriers in non-teaching occupations are muted or amplified by dynamic effects that operate through our model’s human capital accumulation channel.³ The previous literature has explored the static gains and losses extensively in the context of linear models. The key contribution of our paper is (a) the introduction of non-linear wage functions that emphasize the role of absolute advantage in addition to the prior focus on comparative advantage and (b) the intergenerational dynamics of human capital accumulation that mainly operate through the distinct education sector in our model.

In addition to our theoretical contribution, our paper revives and older, but rarely used longitudinal dataset. *Project TALENT* surveyed a representative sample of approximately 377 thousand U.S. high school students in 1959. These same students were interviewed again one, five, and eleven years after they graduated from high school. This dataset was digitized somewhat recently and we use it in combination with the NLSY79 and NLSY97 for our quantitative work. Together, the three surveys paint a rich and detailed picture of American youth spanning more than four decades. Importantly, respondents in all three surveys take aptitude tests that allow us to construct ability profiles, which play a central role in our quantitative work.

Our paper is most closely related, yet complementary, to ?. They focus on the role that rigid employment contracts play in teaching. Similarly, ? explores contract rigidities in the form of licensing requirements, which are a salient feature in the U.S. context. ? explores how more flexible contracts benefitted high-quality teachers in Wisconsin. In our model, the wage function captures some degree of contract rigidity and we can quantify the short and long-run implications of increases contract flexibility. Our paper’s main contribution, however, is a sharp characterization of the tradeoff between static and dynamic aggregate welfare gains by way of a micro-founded human capital accumulation technology embedded in an other standard occupational choice model. The importance of teacher quality has already been documented in ? and ?. Our structural model enables us to identify the forces that shape occupational choice and flexibly characterizes the career choices of prospective teachers with different cognitive abilities over time.

The remainder of the paper is organized as follows. Section 2 introduces the model. Section 3 characterizes the equilibrium. Section 4 reviews the data and we calibrate the model to match

³?, for instance, finds that high-ability math teachers are more likely to exit the teaching profession.

salient empirical moments in section 5. Section 6 concludes.

2 Model

The model is populated by a constant measure M of workers who are employed in one of the I sectors of the economy. Each individual is born with an exogenous I -dimensional vector of skills $\vec{a}$, where each element characterizes her innate ability in any one of the $i \in \{1, \dots, I\}$ occupations in the economy, each corresponding to one of the sectors or industries. This vector of abilities is drawn from a known joint distribution $F(\vec{a})$. Moreover, the measure of workers is partitioned into G different groups, which are indexed by $g \in \{1, \dots, G\}$. Workers are born into one of these groups and membership is permanent. Moreover, we assume that the distribution of abilities is identical across groups, i.e., $F^g(\vec{a}) = F^{g'}(\vec{a}) = F(\vec{a})$.

Without loss of generality, occupation 1 is *teaching*. In this economy, teachers are distinct from other occupations since their technology produces human capital rather than the homogeneous final good in the remaining $I - 1$ sectors. This kind of distinction is salient since we are assuming that the economy is populated by overlapping generations of workers who accumulate human capital when young. When they reach working age (i.e., they are “old”), their human capital contributes toward the efficiency units of labor they supply to the labor market, including home production.

2.1 Preferences and Technologies

As in ?, workers make occupational choices based on their full range of idiosyncratic abilities $\vec{a}$ and the corresponding “payoffs.” To this basic mechanism we add ?-style forces that distort the allocation of skill across occupations and a human capital formation technology with time, good, and teacher inputs.

The timing convention is similar to ?. In the first period, prospective workers – or students – make occupation-specific human capital investments. Since there is no uncertainty, students have perfect foresight and their investment decision will depend on their fully anticipated occupational choice in period 2. They retire at the end of their working period and are replaced by a new cohort of students with measure $\frac{M}{2}$. The environment is stationary in the sense that the members of each cohort draw their ability vectors from the same distribution F_a .

The worker’s utility depends on her consumption and time devoted to human capital investment:

$$U = \max_{C'_g, \{s_{i,g}, e_{i,g}\}_{i=1}^I} \left\{ \mu \ln C'_g + \ln(1 - s_{i,g}) \right\}_{i=1}^I \quad (1)$$

where

$$C'_g = (1 - t')(1 - \tau_{i,g}^{\omega'})\omega'_{i,g}(h_{i,g}) - \sum_{i=1}^I (1 + \tau_{i,g}^e)e_{i,g} \quad (2)$$

$$h'_{i,g} = \kappa(h_{T,g}, a_i)(s_{i,g})^\phi (e_{i,g})^\eta (N(h_{T,g}))^{-\sigma} \quad (3)$$

$$\kappa(h_{T,g}, a_i) = (h_{T,g})^\beta (a_i)^\alpha. \quad (4)$$

C'_g is consumption of the homogeneous final good when old, $s_{i,g}$ is the time allocated to occupation-specific human capital formation when young, and $e_{i,g}$ are units of the final good invested in occupation-specific human capital formation when workers are still students. These investments and the corresponding consumption in the following period are occupation-specific and indexed by the subscript i .⁴ The periodic time endowment is set to unity. $1 - s_{i,g}$ is leisure time when young, and labor is supplied inelastically when individuals reach working age.

Labor income is taxed at a constant marginal rate $t \in [0, 1)$ and $\tau_{i,g}^\omega < 1$ is a group- g -and-occupation- i -specific labor market distortion. We allow for the possibility of *negative* distortions. Finally, $\tau_{i,g}^e$ is a group and occupation-specific educational barrier that raises the cost of good investments in the human capital accumulation technology.

We assume that the draws from $F_a(\cdot)$ are i.i.d. across agents but we allow for the possibility that draws are not independent across occupations.

The production technologies in occupations $2, \dots, I$ are linear in efficiency units of labor $h_{i,g}$ with productivities $\{A_i\}_{i=2}^I$:

$$y_{i,g} = A_i \times h_{i,g}. \quad (5)$$

Since human capital is the only factor of production, we can characterize the income of a worker who is employed in a non-teaching job by:

$$\omega_i(h_{i,g}) = y_{i,g}. \quad (6)$$

Teaching is distinct from other occupations since it “transforms” students with skill endowment $\vec{a}$ into workers with human capital $\vec{h}$ based on equation (3). The acquisition of human capital depends on the teacher’s input $h_{T,g}$, the student’s talent $\vec{a}$, the student’s time and good investments $s_{i,g}$ and $e_{i,g}$, as well as class size $N(h_{T,g})$, which we allow to vary with the teacher’s $h_{T,g}$. Note that students already select an occupation at this stage and the optimal investment depends on a single element, say a_i , of $\vec{a}$.

⁴Nothing prevents students from making investments in human capital in multiple occupations. Since there is no uncertainty in future payoffs, they will, however, select only one in equilibrium.

2.2 Occupational Choice

Individuals will choose the occupation that delivers the highest lifetime utility. The workers' choice between non-teaching occupations is typical in the sense that it is based on a "conventional" ?-type model that has been used extensively in the previous literature. Teaching, on the other hand, is modeled in a way that deviates from the standard assumptions in two key respects: (1) the teachers' human capital is an input into the *students'* human capital production function rather than the production function for final goods and (2) the teacher's wage $\omega_{T,g}(h_{T,g})$ need not be linear in $h_{T,g}$. This implies that the choice of becoming a teacher doesn't just depend on a worker's comparative advantage, it also depends on her *absolute* advantage.

Due to the special nature of teaching, we will discuss the key characteristics based on a version of the model with $I = 2$. This allows us to highlight the novel elements of the model more effectively. In this version, we label the two occupations with subscripts T (for *teaching*) and O (for *other*). Later, and especially when we solve the quantitative version of the model in order to match salient moments in U.S. data, we will re-introduce the additional $I - 2$ occupations.

2.2.1 Teachers and Class Size

In equations (3) and (4), the student's ability in occupation i , denoted a_i , and the teacher's human capital h_T are complements. This implies positive sorting between teachers and students. This effect, however, is tempered by the teacher's span-of-control over the students in her class. $N(\cdot)$ denotes the class size and the coefficient $-\sigma$ captures the stylized fact that in larger classes teachers can pay less attention to one individual student. Here, we allow class size to depend on $h_{T,g}$ and we are focusing on $N(h_{T,g})$ such that – for given $e_{i,g}$ and $s_{i,g}$ – students are indifferent between teachers with different levels of human capital, say $h_{T,g}$ and $h_{T,g}'$:

$$N(h_{T,g}') = \left(\frac{h_{T,g}'}{h_{T,g}} \right)^{\frac{\beta}{\sigma}} N(h_{T,g}). \quad (7)$$

This implies that better teachers work in larger classrooms.

All $\frac{M}{2}$ students in the economy attend school. Therefore, the following resource constraint must be satisfied:

$$\frac{M}{2} = \int_0^\infty N(h_{T,g}) d\mathcal{F}_T(h_{T,g}) \quad (8)$$

where $\mathcal{F}_T(\cdot)$ is the c.d.f. of human capital in teaching.

The resource constraint (8) anchors the class size distribution and equation (7) characterizes the size distribution of classes.

Proposition 1 (Class size distribution) *For a given teacher with human capital $h_{T,g}$, her class size is characterized by:*

$$\left(\frac{2\tilde{H}_T}{M}\right)^\sigma = h_{T,g}^\beta N(h_{T,g})^{-\sigma}, \quad (9)$$

where $\tilde{H}_T = \sum_{g=1}^G \int_0^\infty h_{T,g}^{\frac{\beta}{\sigma}} d\mathcal{F}_T(h_{T,g})$.

2.2.2 Student-Teacher Matching

While this guarantees that students are indifferent between different teachers, it may be the case that better teachers prefer to work with higher ability students. This can happen, for instance, if students make human capital investments in terms of goods as a function of the teacher's human capital $h_{T,g}$. The increased investment then increases the teacher's marginal product, which would be a force for positive sorting.

For simplicity, we assume that students make investments as a function of a and occupation i , but not $h_{T,g}$. The joint assumption that $e_{i,g}(a_i)$ is a function of a (but not $h_{T,g}$) and that $N(h_{T,g})$ is a function of $h_{T,g}$ (but not a_i) implies that the student-teacher matching is random, i.e., the human capital production function is modular in $h_{T,g}$ and a_i .⁵

2.3 Values

The structure we impose on class size and student-teacher matching simplifies the characterization of an equilibrium considerably. While idiosyncratic heterogeneity still matters, the aggregate state can be summarized by a single moment of the distribution of human capital among current teachers (which we discuss in more detail below) rather than the entire distribution of $\{h_{T,g}\}_{g=1}^G$. Thanks to this feature we can characterize stationary equilibria as well as full transition paths sharply.

The idiosyncratic state of a student in this model consists of the I -dimensional vector of abilities $\vec{a}$. Recall that we assume $I = 2$ for the time being, where T denotes teaching and O lumps together all other occupations. An individual teachers' human capital h_T is not part of the

⁵The anecdotal evidence suggests that student-teacher assignments don't involve systematic sorting at the school level. There is, however, evidence in support of sorting between families with highly educated parents and high-ability teachers into better funded schools or school districts. Identifying the empirically relevant extent of this type of sorting is work in progress. Future versions of our model may eventually incorporate it.

idiosyncratic state space. A single moment of the distribution off all teachers' human capital is part of the aggregate state space.

A student in group g solves a maximization problem which entails a discrete occupational choice as well as continuous human capital investments in terms of time and final goods. Formally, she solves:

$$V_g(a_T, a_O, \tilde{H}_T) = \max_{\{s_{O,g}, s_{T,g}, e_{O,g}, e_{T,g}\}} \left\{ V_{O,g}(a_O, \tilde{H}_T), V_{T,g}(a_T, \tilde{H}_T) \right\} \quad (10)$$

where

$$\begin{aligned} V_{O,g}(a_O, \tilde{H}_T) = & \ln \left(1 - s_{O,g} \left(a_O, \tilde{H}_T \right) \right) \\ & + \mu \ln \left[h'_{O,g} A'_O (1 - t') (1 - \tau_{O,g}^{\omega'}) \right. \\ & \left. - e_{O,g}(a_O, \tilde{H}_T) (1 + \tau_{O,g}^e) \right], \end{aligned} \quad (11)$$

$$\begin{aligned} V_{T,g}(a_T, \tilde{H}_T) = & \ln \left(1 - s_{T,g} \left(a_T, \tilde{H}_T \right) \right) \\ & + \mu \ln \left[\omega' (h'_{T,g}, \tilde{H}'_T) (1 - t') (1 - \tau_{T,g}^{\omega'}) \right. \\ & \left. - e_{T,g}(a_T, \tilde{H}_T) (1 + \tau_{T,g}^{e'}) \right] \end{aligned} \quad (12)$$

To make further progress, we need to characterize an occupational choice threshold along which a student with abilities (a^T, a^O) is indifferent between teaching and production work. More formally, the threshold is a function

$$a_{T,g}^*(a_O) = \bar{a}_g(a_O, \tilde{H}_T)$$

such that

$$V_{O,g}(a_O, \tilde{H}_T) = V_{T,g} \left(a_{T,g}^*(a_O), \tilde{H}_T \right), \text{ for all } a_O \in (0, \infty) \quad (13)$$

Unless it's necessary, we will omit the aggregate state argument $\tilde{H}_T$ from $\bar{a}_g(\cdot)$ in order to keep the notation clean and tidy.⁶

⁶We will show later that the occupational threshold is indeed independent from $\tilde{H}_T$. At this point, this result is not (yet) obvious.

The optimization problem is subject to the following constraints:

$$t \left[\sum_{g=1}^G \int_0^\infty \omega_{T,g}(h_{T,g}(a)) f_{T,g}(a) da + \sum_{g=1}^G \int_0^\infty \omega_{O,g}(h_{O,g}(a)) f_{O,g}(a) da \right] = \sum_{g=1}^G \int_0^\infty \omega_{T,g}(h_{T,g}(a)) f_{T,g}(a) da \quad (14)$$

where

$$f_{T,g}(a) = \int_0^{\bar{a}_g^{-1}(a)} f(a, b) db, \quad (15)$$

$$f_{O,g}(b) = \int_0^{\bar{a}_g(b)} f(a, b) da. \quad (16)$$

$f(a, b)$ is the p.d.f. associated with the c.d.f. $F(\cdot)$ from section 2.1 in the special case with $I = 2$, $f_{T,g}(a)$ describes today's distribution of teachers, and $f_{O,g}(a)$ characterizes the distribution of other workers. Workers have perfect foresight and correctly anticipate next period's tax rate t' and future labor market distortions $\left\{ \tau_{O,g}'^{\omega}, \tau_{T,g}'^{\omega} \right\}_{g=1}^G$ that satisfy (14)-(16) one period from today.⁷

The aggregate laws of motion for human capital in teaching ($\tilde{H}_T$) and "other" (H_O) are:

$$H_O' = \sum_{g=1}^G \int_0^\infty \left(\frac{2\tilde{H}_T}{M} \right)^\sigma a^\alpha s_{O,g} \left(a, \tilde{H}_T \right)^\phi e_{O,g}(a, \tilde{H}_T)^\eta f_{O,g}(a) da \quad (19)$$

$$\tilde{H}_T' = \sum_{g=1}^G \int_0^\infty \left(\left(\frac{2\tilde{H}_T}{M} \right)^\sigma a^\alpha s_{T,g} \left(a, \tilde{H}_T \right)^\phi e_{T,g}(a, \tilde{H}_T)^\eta \right)^{\frac{\beta}{\sigma}} f_{T,g}(a) da \quad (20)$$

Teachers are public sector employees and their compensation is funded by tax revenues. Equation (14) is the (local) government's balanced budget requirement, which is a salient feature in the U.S. context.⁸

⁷In the more general case with $I > 2$, $i = 1$ indexes teachers and the measures of teachers and other workers from equations (15) and (16) are given by:

$$f_{T,g}(a_T) = \sum_{i=2}^I \int_0^{\bar{a}_{i,g}^{-1}(a_T)} \Pi_{j \neq i} F \left(\left[\frac{A_i}{A_j} \cdot \frac{1 - \tau_{i,g}'^{\omega}}{1 - \tau_{j,g}'^{\omega}} \cdot \left(\frac{1 + \tau_{i,g}^e}{1 + \tau_{j,g}^e} \right)^{-\eta} \right]^{\frac{1}{\alpha}} \cdot a_i \right) f_{i,g}(a) da \text{ and} \quad (17)$$

$$f_{i,g}(a_i) = F(\bar{a}_{i,g}(a_i)) \Pi_{j \neq i} F \left(\left[\frac{A_i}{A_j} \cdot \frac{1 - \tau_{i,g}'^{\omega}}{1 - \tau_{j,g}'^{\omega}} \cdot \left(\frac{1 + \tau_{i,g}^e}{1 + \tau_{j,g}^e} \right)^{-\eta} \right]^{\frac{1}{\alpha}} \cdot a_i \right), \text{ respectively.} \quad (18)$$

⁸The current version of the model has a single location. Arguably, a more salient environment would feature multiple locations, each of which is subject to a local balanced budget requirement. This extension introduces

Equation (19) is the aggregation over h_O of the current cohort of students, while equation (20) is the $\left(\frac{\beta}{\sigma}\right)^{\text{th}}$ moment of the distribution of h_T of the current cohort of students.

Note that H'_O depends on $\tilde{H}'_T$ but not on H_O . This reflects the assumption that the acquisition of human capital by today's students – regardless of the occupational choice – depends on the quality of today's teachers but not on the human capital of today's production workers.

The teachers' wage profile ω_T is exogenous. We assume that ω_T is strictly increasing in the teacher's human capital h_T and continuously differentiable. We allow, in particular, for the possibility that ω_T is not linear in h_T .

3 Equilibrium

Definition 1 *Given next period's tax rate t' , the labor market distortions and educational barriers $\left\{ \left\{ \tau_{i,g}^e, \tau_{i,g}^{\omega'} \right\}_{i=1}^I \right\}_{g=1}^G$, the occupational choices of today's old agents and aggregate human capital $\tilde{H}_T$ and H_O , the equilibrium consists of the individual investment decisions of young agents, $\{e_{T,g}, s_{T,g}, e_{O,g}, s_{O,g}\}$, the occupational choice boundary $a_{T,g}^*(a_O)$, the corresponding densities $f_{T,g}(\cdot)$ and $f_{O,g}(\cdot)$, and occupation- and group-specific wage profiles $\{\omega_{T,g}, \omega_{O,g}\}$ such that:*

1. *Given the wage profiles, each individual's choice of goods investment $\{e_{T,g}, e_{O,g}\}$ and time investment $\{s_{T,g}, s_{O,g}\}$ maximizes (10).*
2. *The aggregate human capital stocks follow the laws of motion given by (19) and (20).*
3. *The government's budget constraint (14) is satisfied.*

3.1 Human Capital Investment

Let's first characterize the optimal human capital investment for a prospective teacher with ability a_T from group g .

The F.O.C.s for $s_{T,g}$ and $e_{T,g}$, respectively, after a few steps of algebra are:

$$(1 + \tau_{T,g}^e) e_{T,g} \left(\frac{\sigma}{\eta\beta} - 1 \right) = (1 - s_{T,g}) \mu (1 - t') (1 - \tau_{T,g}^{\omega'}) \cdot \frac{\partial \omega'_{T,g}}{\partial h'_{T,g}} \cdot \frac{\partial h'_{T,g}}{\partial s_{T,g}} \quad (21)$$

$$1 + \tau_{T,g}^e = (1 - t') (1 - \tau_{T,g}^{\omega'}) \cdot \frac{\partial \omega'_{T,g}}{\partial h'_{T,g}} \cdot \frac{\partial h'_{T,g}}{\partial e_{T,g}}, \quad (22)$$

additional complexities and we focus on this simple version for now.

Let $\tau_{T,g} \equiv \frac{(1-t')(1-\tau_{T,g}^{\omega'})}{1+\tau_{T,g}^e}$ be a composite wedge that takes into account the rate at which future income is taxed (t'), the extent of labor market discrimination ($\tau_{T,g}^{\omega'}$), and educational barriers ($\tau_{T,g}^e$). Note that $\tau_{T,g} = 1$ when $t' = \tau_{T,g}^{\omega'} = \tau_{T,g}^e = 0$.

The amount of time and goods this individual invests in human capital is then given by:

$$s_{T,g} = \frac{\mu\phi}{\mu\phi + \eta \cdot \tau_{T,g} \cdot \frac{\omega'_{T,g}(h'_{T,g})}{e_{T,g}} - \eta} \quad (23)$$

$$e_{T,g} = \left(\tau_{T,g} \cdot \eta \cdot \left(\frac{2\tilde{H}_T}{M} \right)^\sigma \cdot \left(\frac{\partial \omega'_{T,g}}{\partial h'_{T,g}} \right) \cdot s_{T,g}^\phi \cdot a_T^\alpha \right)^{\frac{1}{1-\eta}} \quad (24)$$

The F.O.C.s for prospective production workers (i.e., those labeled “other”) are analogues of (23) and (24) with the simplification that the wage profile in production (as opposed to teaching) is linear in $h_{O,g}$ and the optimal amount of time and resources invested in human capital is given by:

$$s_{O,g} = \frac{\mu\phi}{\mu\phi + 1 - \eta} \quad (25)$$

$$e_{O,g} = \left(\tau_{O,g} \cdot \eta \cdot \left(\frac{2\tilde{H}_T}{M} \right)^\sigma \cdot A'_O \cdot s_{O,g}^\phi \cdot a_O^\alpha \right)^{\frac{1}{1-\eta}} \quad (26)$$

where, analogously to the problem of a prospective teacher, we define $\tau_{O,g} \equiv \frac{(1-t')(1-\tau_{O,g}^{\omega'})}{1+\tau_{O,g}^e}$.

3.2 The Aggregate Laws of Motion

After substituting (23)-(26) into (19) and (20) we can characterize the equilibrium laws of motion for $\tilde{H}_T$ and H_O by:

$$\tilde{H}'_T = \eta^{\frac{\beta\eta}{\sigma(1-\eta)}} \cdot \left(\frac{2\tilde{H}_T}{M} \right)^{\frac{\beta}{1-\eta}} \cdot \sum_{g=1}^G \tau_{T,g}^{\frac{\beta\eta}{\sigma(1-\eta)}} \cdot \int_0^\infty \left(\frac{\partial \omega'_{T,g}}{\partial h'_{T,g}} \right)^{\frac{\beta\eta}{\sigma(1-\eta)}} \cdot s_{T,g}^{\frac{\beta\phi}{\sigma(1-\eta)}} \cdot a^{\frac{\beta\alpha}{\sigma(1-\eta)}} f_{T,g}(a) da \quad (27)$$

and

$$H'_O = \eta^{\frac{\eta}{1-\eta}} \cdot \left(\frac{2\tilde{H}_T}{M} \right)^{\frac{\sigma}{1-\eta}} \cdot \sum_{g=1}^G \tau_{O,g}^{\frac{\eta}{1-\eta}} \cdot \int_0^\infty A'^{\frac{\eta}{1-\eta}}_O \cdot s_{O,g}^{\frac{\phi}{1-\eta}} \cdot a^{\frac{\alpha}{1-\eta}} f_{O,g}(a) da. \quad (28)$$

We can show that this economy has a steady state in $\tilde{H}_T$ and we can characterize it implicitly in a single equation:

$$\tilde{H}_T = \eta^{\frac{\beta\eta}{\sigma(1-\eta-\beta)}} \cdot \left(\frac{2}{M}\right)^{\frac{\beta}{1-\eta-\beta}} \cdot \left[\sum_{g=1}^G \tau_{T,g}^{\frac{\beta\eta}{\sigma(1-\eta)}} \cdot \int_0^\infty \left(\frac{\partial\omega_{T,g}}{\partial h_{T,g}}\right)^{\frac{\beta\eta}{\sigma(1-\eta)}} \cdot s_{T,g}^{\frac{\beta\phi}{\sigma(1-\eta)}} \cdot a^{\frac{\beta\alpha}{\sigma(1-\eta)}} f_{T,g}(a) da \right]^{\frac{1-\eta}{1-\eta-\beta}} \quad (29)$$

3.3 Occupational Choice

Proposition 2 (Occupational Choice Boundary) *The occupational choice boundary, denoted $\bar{a}_{T,g}(a_O)$, depends on the magnitude of the educational barriers, the labor market frictions, and the student's teaching-specific human capital $h'_{T,g}$. It does not, however, depend on either current or future aggregate human capital in teaching $\tilde{H}_T$. It is characterized by:*

$$\left(\frac{a_T}{a_O}\right)^\alpha \cdot \frac{\tau_{T,g}}{\tau_{O,g}} \cdot \left(\frac{1 + \tau_{T,g}^e}{1 + \tau_{O,g}^e}\right)^{1-\eta} \cdot \left(\frac{1 - s_{T,g}}{1 - s_{O,g}}\right)^{\frac{1-\eta}{\mu}} \cdot \left(\frac{s_{T,g}}{s_{O,g}}\right)^\phi \cdot \frac{\frac{\partial\omega'_{T,g}}{\partial h'_{T,g}}}{A'_O} \cdot \left(\frac{\frac{\omega'_{T,g}}{\frac{\partial\omega'_{T,g}}{\partial h'_{T,g}} \cdot h'_{T,g}} - \eta}{1 - \eta}\right)^{1-\eta} = 1. \quad (30)$$

4 Data

We parameterize the model with data from longitudinal surveys that cover American youth at three different times after World War II: Project TALENT, NLSY 79, and NLSY 97. While the two NSLY surveys have been used extensively in the literature, Project TALENT is somewhat less known and we describe it in some detail below.

4.1 Project TALENT

Project TALENT surveyed approximately 377,000 students enrolled in high school in 1960. The vast majority of these students were in the 14-18 year age range with a few outliers outside this bracket (on both sides). The sample of surveyed students was representative and accounted for 5 percent of the high school population that year. The project was funded by the United States Office of Education (the precursor of the Department of Education) and its goal was to collect high quality data on the achievements, aptitudes, and interests of high

school students in the United States, and to examine whether, and how, these data predicted future outcomes in terms of educational attainment, career, and wellbeing.

In the initial wave of surveys, students were subjected to a two-day battery of aptitude, ability, and achievement tests. In addition, students answered questions about dispositional traits, their interests (with respect to occupations and activities), as well as personal and family information. In total, the surveyed students answered more than one thousand questions.

Follow-up surveys were then carried out one, five, and eleven years after the students' expected high school graduation. For instance, students who were in eleventh grade in 1960 were expected to graduate in 1961 and were contacted again in 1962, 1966, and 1972. Ninth graders were contacted in 1963, 1967, and 1973. A lack of federal funding stopped the project during the planning stage for a 17-year follow-up, which never took place.

51 percent of the original participants participated in the 1-year post-graduation surveys. The five and eleven-year follow-ups successfully contacted 35 and 25 percent of the original participants, respectively. One concern with attrition rates of this magnitude is that the survey sample of waves two through five is subject to selection along one or several attributes.

In Table 1 we summarize the descriptive statistics by gender, race, parental education, and parental economic status (all in 1960). The racial composition of the sample exhibits some anomalies. The racial classification in the 1960 base year is problematic since almost 60 percent of the sample are in the "other", "unknown", or "conflicting" category.⁹

Although the high attrition rate from the original sample size of almost 370,000 is a concern, Table 1 suggests that selection along the respondents' gender and their parents' educational attainment and income is not a major issue. The 11-year post survey appears to be roughly representative of the high school population in 1960. On average, the respondents had slightly more educated parents and higher household income in 1960. This is, arguably, far from ideal but selection is not a major concern at this point.

The rationale for using panel data from Project TALENT and the two NLSYs is that all three surveys allow us to connect scores from aptitude tests to labor market outcomes. Project TALENT contains a number of tests administered in 1960 when the respondents were attending high school, though in different grades.¹⁰ We mainly use composite scores built from a battery of underlying tests measuring verbal and mathematical aptitude as well as social skills. The first includes literature information, vocabulary, and English. The second contains math information, arithmetic reasoning, introductory high school math, and advanced high school

⁹The classification of respondents into race categories in the base year suffers from inconsistencies (e.g. self-reported vs. pollster's classification). Moreover, the sample proportion of black high school students differs from that reported in the 1960 Census.

¹⁰Roughly one quarter of the sample was in each high school grade (i.e. 9 through 12) in the baseline survey.

	Base Year	1-Year Post	5-Year Post	11-Year Post
Sample Size	368,938	188,138	128,163	92,962
Black (% of sample size)	6,488 (1.8%)	4,003 (2.1%)	4,856 (3.8%)	3,026 (3.3%)
Other / Unknown / Conflicting (% of sample size)	215,786 (58.5%)	73,668 (39.1%)	3,739 (2.9%)	3,758 (4.0%)
Female (% of sample size)	184,912 (50.1%)	98,183 (52.2%)	64,841 (50.6%)	47,417 (51.0%)
Parents: High School Degree (% of sample size)	155,128 (42.05%)	85,804 (45.61%)	60,142 (46.93%)	43,204 (46.47%)
Parents: College Degree (% of sample size)	58,959 (15.98%)	35,065 (18.64%)	25,193 (19.66%)	17,361 (18.68%)
Parents: High Income (% of sample size)	116,628 (33.42%)	65,691 (36.70%)	46,499 (38.10%)	33,097 (37.43%)

Table 1: Project TALENT Demographic Composition

math. Table 2 shows summary statistics for the 88,330 individuals who participated in the 11-year follow-up survey. We focus on the 11-year *post* wave since it contains the bulk of occupational information at a time when the respondents are approximately thirty years old.¹¹

In the table we report gender and composite aptitude scores for mathematics, verbal and social skills, split between teachers and workers in other occupations as well as the unemployed and those not in the labor force.

Approximately 60 percent of teachers in the sample are female eleven years after their presumed high school graduation. In non-teaching occupations the share of women is 30 percent and women account for more than 90 percent of respondents who report that they are not in the labor force.

On average, teachers have higher mathematics, verbal, and social test scores than respondents in other occupations and those who are not in the labor force. Together, these stylized facts are consistent with the hypothesis that high-ability women are disproportionately choosing teaching jobs and careers and we will return to this point below.

If the labor market discriminated more against certain groups (gender, race,...) in non-teaching occupations in 1960, we expect these groups to favor teaching over other occupations or to stay out of the labor force altogether. “Not in the labor force” is more a reflection of the absolute level of labor market discrimination against a particular group while the oc-

¹¹The 5-year post-graduation survey contains occupational information but we have not made use of it yet.

	11-Year Post				
	Teachers	Other	Unemployed	Not in Labor Force	Total
Female (% of subsample)	3,641 (60.36%)	16,063 (30.47%)	1,154 (61.84%)	23,001 (92.84%)	43,859 (51.36%)
Math composite mean (st.dev.)	2.964 (0.888)	2.704 (1.023)	2.573 (1.052)	2.477 (0.939)	2.654 (1.0)
Verbal composite mean (st.dev.)	6.370 (0.809)	5.900 (1.031)	5.909 (1.145)	6.091 (0.929)	5.989 (1.0)
Social composite mean (st.dev.)	2.915 (0.988)	2.660 (1.000)	2.563 (0.985)	2.753 (0.996)	2.703 (1.0)
IQ composite mean (st.dev.)	4.330 (0.822)	3.964 (1.027)	3.887 (1.128)	3.978 (0.955)	3.992 (1.0)
Observations (% of total sample)	6,032 (7.06%)	52,726 (61.74%)	1,866 (2.19%)	24,775 (29.01%)	85,399 (100.0%)

Table 2: Project TALENT Test Scores by Occupation

occupational choice between teaching or other occupations reflects differences between group-specific barriers. To the extent that occupations require specific human capital investments we allow for educational barriers to affect occupational choices separately in the model. If both of these barriers declined after 1960, we would expect a narrowing of the educational achievement gap and more talented females should move from teaching into other occupations. This reallocation would lower the average quality of teachers and, therefore, we expect to observe more significant differences in the ability distributions between teachers and other occupations in cohorts who entered the labor market in the 60s and 70s (i.e., our Project TALENT population) compared to younger individuals (i.e., the NLSY79 and NLSY97 respondents).

We focus on the verbal and math composite scores provided by Project TALENT. In addition, we construct a measure of non-cognitive ability, which we call social ability, by combining scores on impulsiveness, calmness, self-confidence, and personality maturity by way of a principal component analysis (PCA). Finally, we convert these scores into percentile ranks among individuals. The conversion to percentiles enables us to connect the cognitive assessments in Project TALENT to the ASVAB and AFQT tests in the two NLSYs.¹²

Figure 1 shows the distribution of math and verbal composite scores by occupations in Project TALENT. The distribution of math scores in panel 1a is skewed to the right for “other” compared to the distribution of math scores among teachers. Panel 1b shows a similar pattern in

¹²The percentiles are not reported here. They are available by request.

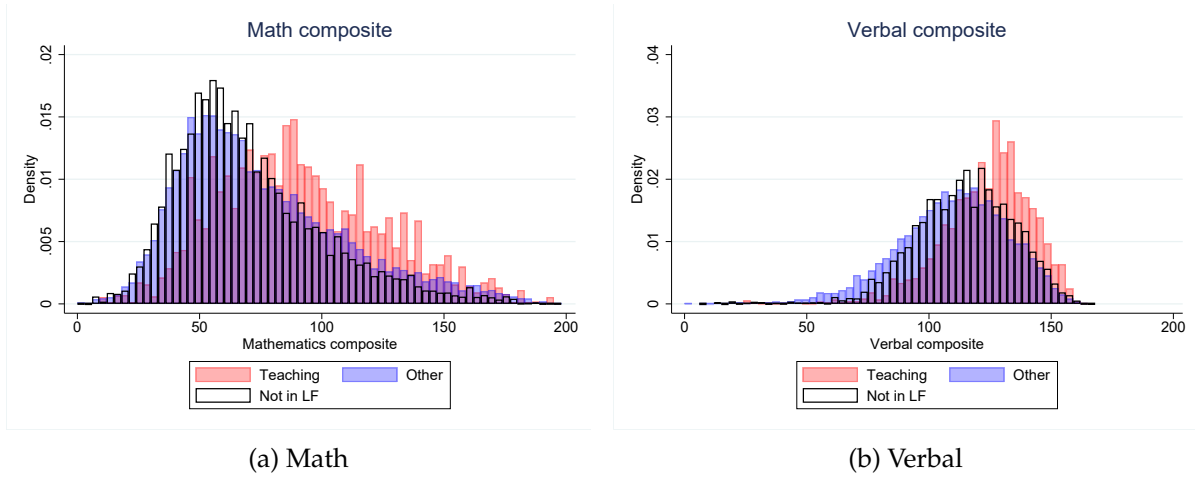


Figure 1: Composite Scores by Occupation in Project TALENT

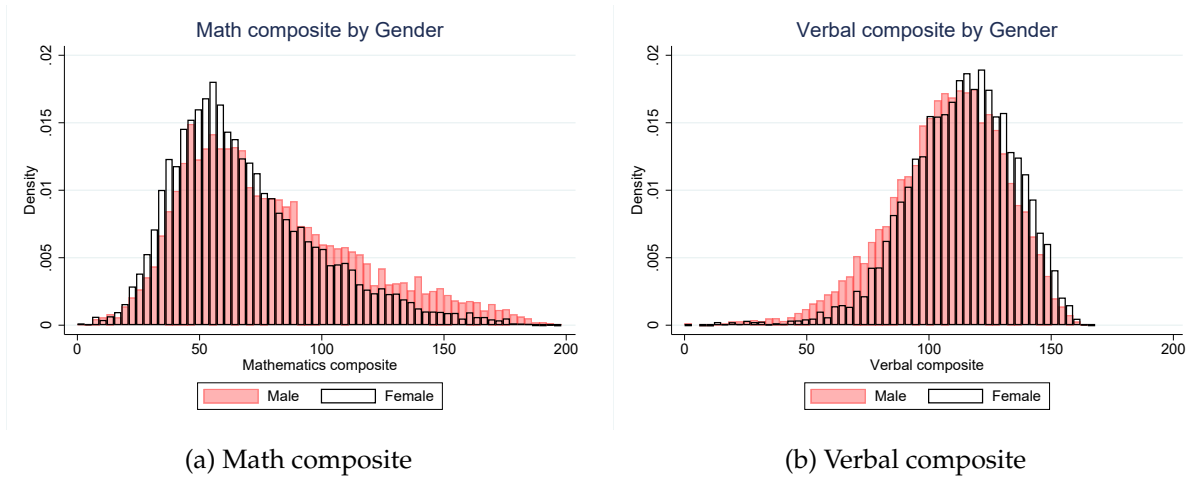


Figure 2: Composite Scores by Gender in Project TALENT

terms of relative skewness. The distribution of scores of respondents in non-teaching occupations and those who report not being in the labor force are quite similar. Overall, individuals in teaching occupations tend to have higher scores than non-teachers. One possible explanation for these differences is that individuals select into teaching along a non-skill dimension and an obvious selection candidate is gender. If women indeed exhibited a distinct distribution of aptitude scores, this may explain the discrepancy between teaching and other occupations. The data, however, do not suggest that women are innately more or less talented than men. Women's verbal scores slightly dominate the men's while the opposite is true for the math scores (see Figures 2a and 2b).

The distributions of composite IQ scores for Project TALENT respondents split by gender are quite similar (see Figure 3). Figures 1-3 suggest that high-ability women gravitate to-

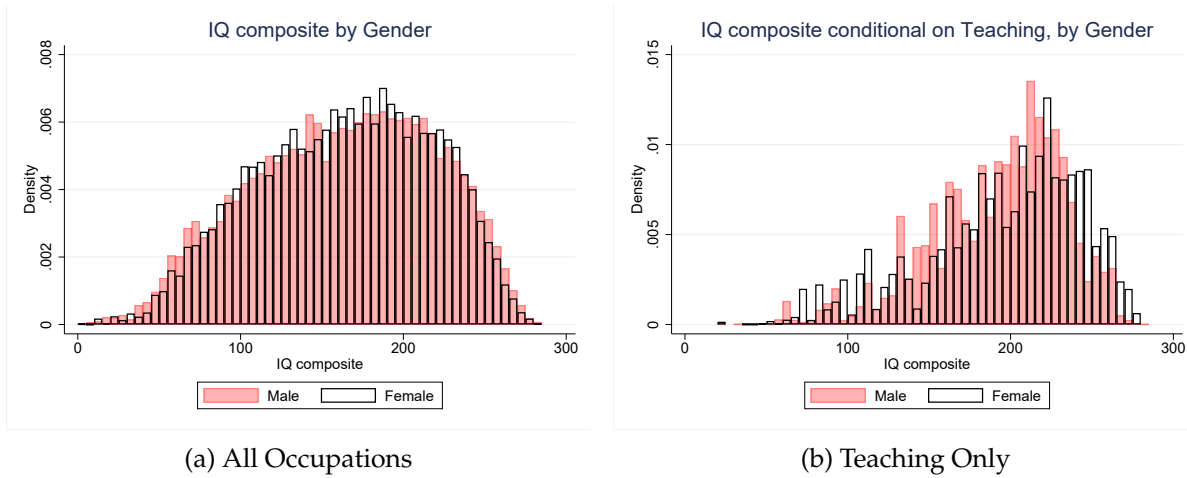


Figure 3: Composite IQ Scores by Gender in Project TALENT

ward teaching, which is illustrated by the distribution of IQ scores in Figure 3b. Due to the smaller numbers of individuals in that figure compared to the three previous figures (the number of men is particularly low), the plot contains more noise and one should draw any conclusion with appropriate caution. With this in mind, it does appear that teaching attracts higher-ability individuals and that the selection of women along this dimension is somewhat stronger, plausibly due to the lack of career opportunities in other occupations.

4.2 NLSY79 and NLSY97

In contrast to Project TALENT, the two NLSYs initiated in 1979 and 1997 are widely used and many readers are already familiar with the data. ¹³, among many others, provides a comprehensive overview of the two surveys. For the purposes of our paper, the *Armed Forces Qualification Test* (AFQT) scores are of particular interest since we need proxies to capture differences in the individuals' raw skills, which is the crucial source of exogenous variation in our model. The score consists of four sections of the *Armed Services Vocational Aptitude Battery* (ASVAB): arithmetic reasoning, word knowledge, paragraph comprehension, and numerical operations. The tests took place in 1980 and 1999 for NLSY79 and NLSY97 cohorts, respectively. Since the tests were administered at different ages (and times) and used different formats, NLS program staff re-normed the scores in 2006, in order to facilitate cross-cohort comparisons.¹³ We reduce four ASVAB categories into two composite dimensions, verbal and math, by applying a principal component analysis (PCA). The verbal score is the first princi-

¹³ ¹³ also propose a method to enable a comparison of ASVAB scores across the two NLSY samples. Note that we cannot construct equivalence scores that would allow us to compare results from aptitude tests in Project TALENT with those from the two NLSYs.

pal component of word knowledge and paragraph comprehension, and the math score is that of math knowledge and arithmetic reasoning. Next, we combine the *Rotter Locus of Control Scale* and *Rosenberg Self-Esteem Scale* from NLSY79 to construct a measure of non-cognitive ability (??), which we will call social ability hereafter. Likewise, we combine the Big 5 personality factor conscientiousness traits to construct a comparable social ability measure for NLSY97 (?). Finally, we convert these scores into percentile ranks among individuals.¹⁴

Table 3 shows summary statistics for the individuals in our sample, divided into three categories: (1) teachers, (2) non-teachers/others, (3) not in labor force. Since an individual usually has multiple employment spells, she might be in all these categories over the course of the panel. We define an individual as a teacher if she has at least one spell in teaching that is longer than 9 months (one academic year). An individual is out of labor force if she reported to be out of labor force for 208 consecutive weeks for a reason other than schooling.¹⁵

If indeed labor market barriers are shrinking over time, we would expect to see some evidence of this in the occupational choice of women in more recent cohorts and hence in the NLSY79 and NLSY97. In these younger cohorts, the share of women among teachers continues to be high. The gender composition of teachers in NLSY 79 is comparable to the one we observe in Project TALENT. In the more recent NLSY 97, the share is a few percentage points lower. In contrast, the share of female workers in other occupations is rising, from approximately 39 percent in Project TALENT to 44 percent in NLSY79, and 45 percent in NLSY97. The gender composition of those who are out of the labor force is heading in the opposite direction. The share of females dropped to 66 percent among NLSY79 respondents and 64 percent among NLSY97 respondents, compared to almost 80 percent in the 11-year post graduation survey from Project TALENT.

One significant difference between the Project TALENT and NLSY respondents are the aptitude scores for those who are not in the labor force. While their average score was higher than the average score of labor force participants in non-teaching occupations in Project TALENT, the ranking is reversed in the two NLSYs. This suggests that the exit rate from “not in the labor force” is correlated with the respondents’ ability. High and low ability individuals appear to respond differentially to changes in labor market discrimination and this is further corroborated in a set of plots we present below.

Figure 4 plots the NLSY analogues of the math and verbal scores in Project TALENT shown in Figure 1. Clearly, teaching still attracts higher-ability individuals than “other” occupations or those who are not in the labor force. While the ability distribution of teachers still first-order dominates the scores in “other” occupations in the 1979 survey, there is a remarkable shift in

¹⁴Not reported in the current version of the manuscript

¹⁵After four years, “not in the labor force” is effectively an absorbing state. The probability of returning to the labor force is less than 1%.

Panel A: NLSY79					
	Teachers	Other	Unemployed	Not in Labor Force	Total
Female (% of subsample)	999 (43.19%)	2,933 (44.85%)	221 (41.39%)	1,568 (76.34%)	5,721 (50.01%)
Math ability (st.dev.)	2.450 (0.932)	2.443 (1.018)	1.919 (0.935)	1.962 (0.912)	2.334 (1)
Verbal ability (st.dev.)	3.614 (0.864)	3.509 (0.971)	2.920 (1.088)	3.057 (1.074)	3.422 (1)
Social ability (st.dev.)	3.891 (1.016)	3.824 (0.980)	3.566 (0.969)	3.552 (1.011)	3.777 (1)
AFQT (st.dev.)	1.780 (0.928)	1.741 (1.004)	1.184 (0.950)	1.270 (0.956)	1.638 (1)
Observations (% of total sample)	2,313 (20.22%)	6,539 (57.16%)	534 (4.67%)	2,054 (17.95%)	11,440 (100.00%)
Panel B: NLSY97					
	Teachers	Other	Unemployed	Not in Labor Force	Total
Female (% of subsample)	478 (48.53%)	2,184 (46.60%)	171 (48.03%)	605 (64.71%)	3,438 (49.38%)
Math ability (st.dev.)	1.309 (0.885)	1.422 (0.959)	1.710 (1.148)	1.780 (1.164)	1.469 (1)
Verbal ability (st.dev.)	1.475 (0.879)	1.590 (0.957)	1.985 (1.144)	2.001 (1.154)	1.649 (1)
Social ability (st.dev.)	5.684 (0.904)	5.645 (0.967)	5.478 (1.108)	5.460 (1.158)	5.614 (1)
AFQT (st.dev.)	1.776 (0.943)	1.665 (0.998)	1.045 (0.808)	1.185 (0.961)	1.585 (1)
Observations (% of total sample)	985 (14.15%)	4,687 (67.31%)	356 (5.11%)	935 (13.43%)	6,963 (100.00%)

Table 3: NLSY Scores by Occupation

the talent distribution of those who are not in the labor force. It is for more right-skewed than the distribution of workers in “other” occupations. Unfortunately, the 1997 data is considerably more noisy mostly due to the smaller sample size and we’re refraining from reading too much into any particular feature of the data. This being said, it is worth highlighting that the

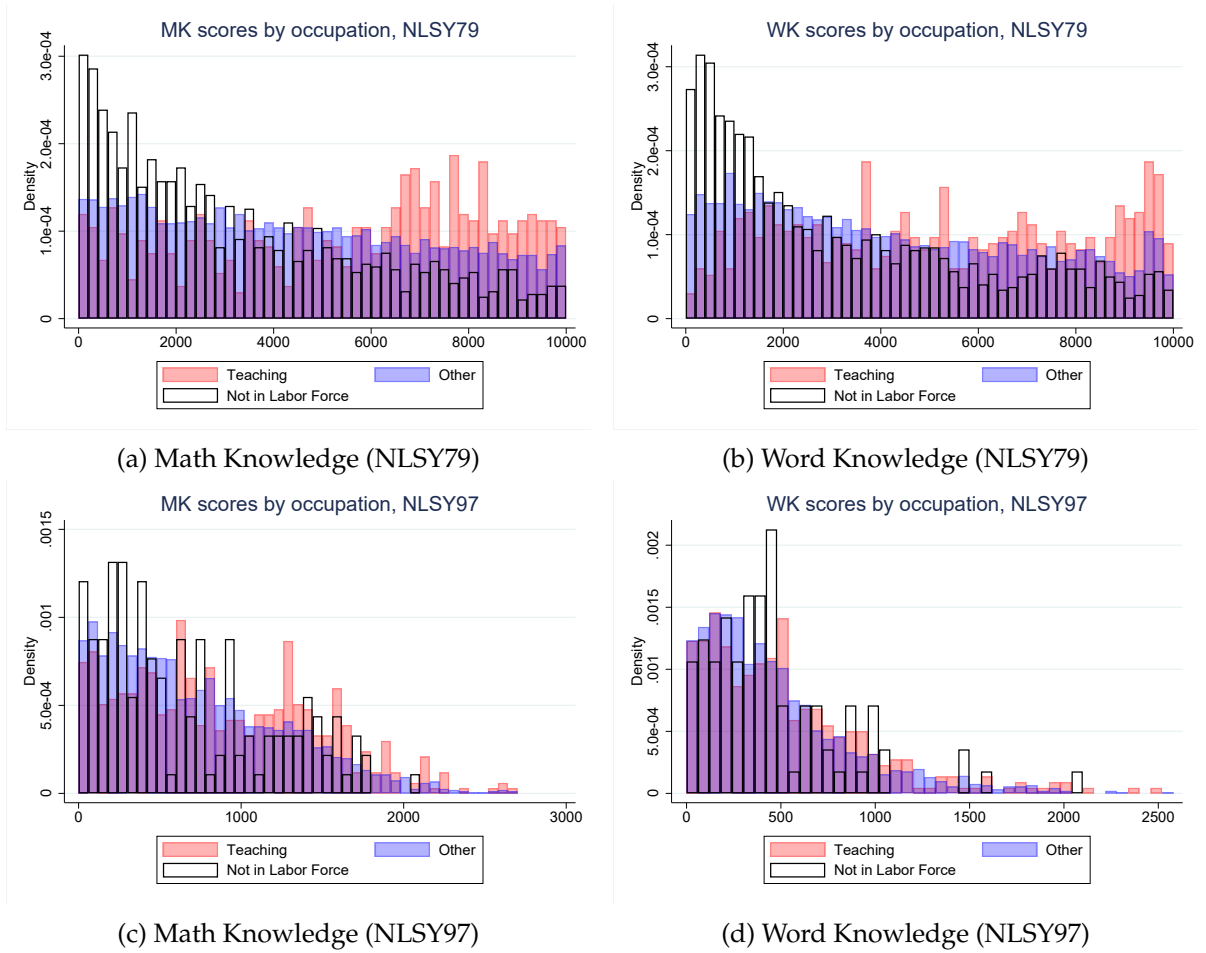


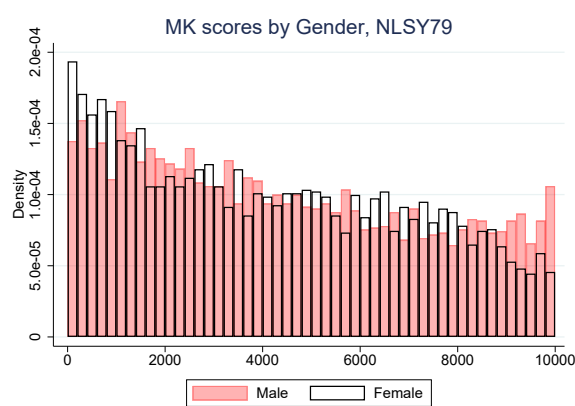
Figure 4: AFQT Scores by Occupations

definition of “not in the labor force” is *not* identical in the TALENT data on the one hand and the NLSY data on the other.¹⁶ It is, however, suggestive of a trend whereby high-ability individuals are pulled into the labor force more “aggressively” compared to low-skill workers over time.

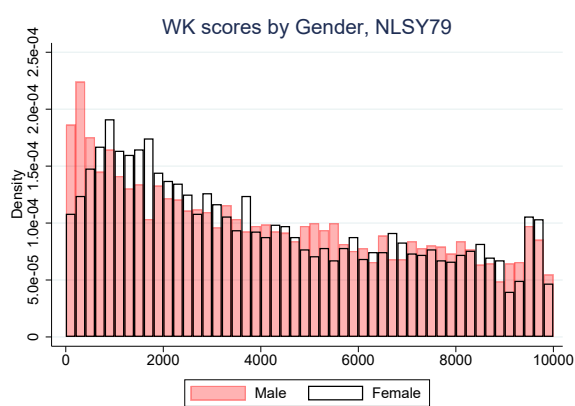
Figures 5 and 6 are the NLSY analogues of the math and verbal composite scores in Figures 2 and 3.

Unfortunately, the composite score distributions aren’t comparable between Project TALENT and NLSY cohorts for several reasons. Although the ability tests try to capture math and verbal abilities, the tests themselves are very different. Therefore, it is impossible to compare changes in score distributions over time. Despite this limitation we can still compare the composition (by gender or occupation) *within* cohorts across the three surveys. In fact, the

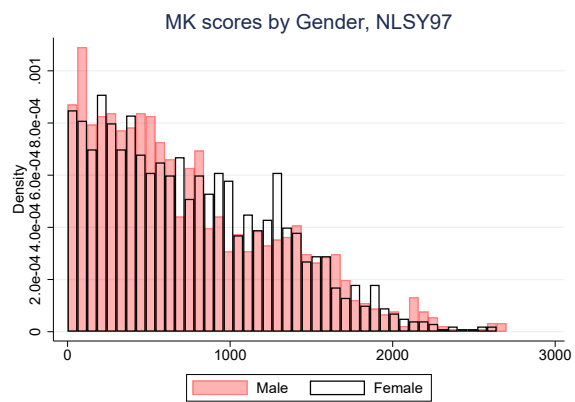
¹⁶The difference stems from the fact that we observe Project TALENT participants relatively infrequently and the respondents’ labor market histories are less complete.



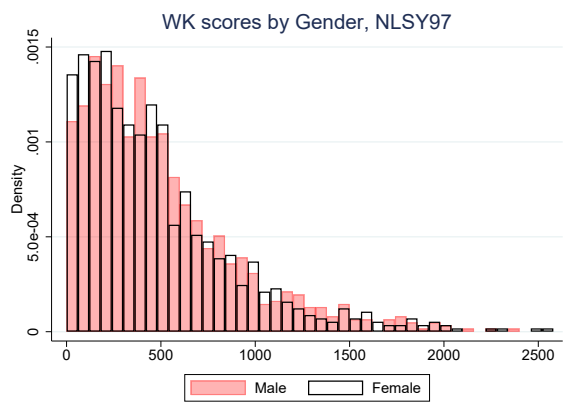
(a) Math Knowledge (NLSY79)



(b) Word Knowledge (NLSY79)

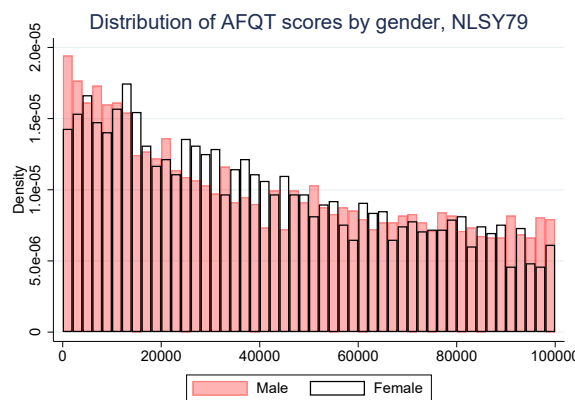


(c) Math Knowledge (NLSY97)

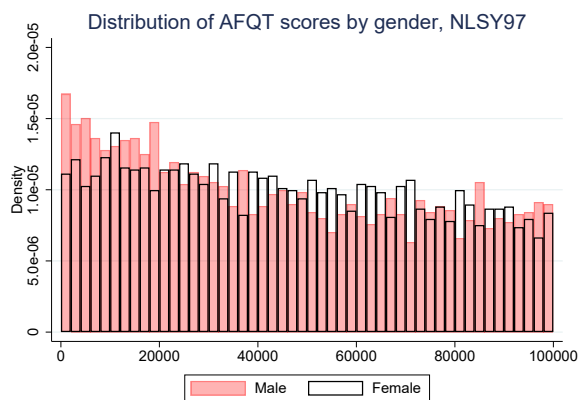


(d) Word Knowledge (NLSY97)

Figure 5: Topic Scores by Gender



(a) AFQT Composite (NLSY79)



(b) AFQT Composite (NLSY97)

Figure 6: AFQT Scores by Gender

particular *Roy* flavor of our model suggests that this is where the interesting action is.

Differences in the frequency of the surveys further complicate comparisons between Project TALENT and the NLSY's. The NLSY79 and NLSY97 survey the respondents far more frequently and this enables us to construct more detailed employment histories and occupation choices over the course of panels. While Project TALENT provides data on the respondents' *current* occupation in the 1, 5, and 11-year post waves, it misses some of the higher-frequency churn since it doesn't ask any retrospective career questions. Moreover, the probability of being a teacher one year after high school graduation is vanishingly small (0.08% of the sample). In the figures and tables presented so far we report the occupations reported in the 11-year follow-up study. While certainly less complete than the NLSY data, Project TALENT still offers insights into the career decisions of young adults at a time for which there is only a very limited supply of nationally representative longitudinal survey data.¹⁷

Project TALENT, NLSY79, and NLSY97 provide us with the data moments that we match to model-generated moments in order to quantify the static and dynamic effects of the implied educational barriers and labor market barriers. While the discussion so far emphasized the contrast between men and women, it is worth emphasizing that our model framework can accommodate more richness in terms of the salient groups of workers. In ongoing work we are incorporating race and ethnicity to more fully capture the salient educational and labor market frictions in the data.¹⁸

4.3 O*NET

The final element of our calibration is based on the U.S. Department of Labor's characterization of occupation-specific skill requirements in O*NET. The data contain the mix of knowledge, skills, and abilities used to perform the tasks that characterize an occupation. For our purposes, we need to convert these descriptors into a three-dimensional job characterization consisting of mathematics, verbal, and social skills. This procedure largely follows ?. First, we use 26 of these descriptors that are most related to the ASVAB component tests and another 6 descriptors related to the social skills and convert them into four ASVAB test categories. Second, we normalize each dimension's standard deviation to be one and reduce these 4 ASVAB categories into 2 composite dimensions, verbal and math, by applying principal com-

¹⁷In the work we have done so far, we impose some of the data limitations in Project TALENT on the NLSY in order to get a better sense for underlying changes in labor market conditions and career decisions across cohorts. In particular, we are only using the occupational choice of 29-year olds in all three surveys even though the NLSYs allow us to construct far more detailed occupational histories. Evidently we're giving up some valuable details in the process for the benefit of more cross-survey comparability.

¹⁸Since the racial categorization is somewhat problematic in Project TALENT, this extension poses additional challenges.

ponent analysis (PCA). The verbal score is the first principal component of word knowledge and paragraph comprehension, and the math score is the first principal component of math knowledge and arithmetic reasoning. All four scores were converted into percentile ranks among occupations. Likewise, we create an index of social ability using the first principal component based on the O*NET descriptors related to social skills. Lastly, we group the full set of O*NET occupations into 20 groups. The 21st occupational group is home production. We collapse the individual occupations into these larger groups by calculating the simple (unweighted) average skill requirement (in terms of percentile ranks).

Last but not least, we use these skill profiles to project the survey respondents' three-dimensional abilities into an I -dimensional vector of job-specific abilities. We decompose the job-specific ability into an absolute and a comparative dimension in a way that aligns naturally with the notion of absolute and comparative advantage. We characterize the comparative dimension for each worker j with a vector of abilities (a_v^j, a_m^j, a_s^j) in each occupation group i with skill requirement (s_v^i, s_m^i, s_s^i) :

$$CA_{i,j} = \frac{a_v^j}{s_v^i} \cdot \left| \frac{a_v^j}{a_v^j + a_m^j + a_s^j} - \frac{s_v^i}{s_v^i + s_m^i + s_s^i} \right| +$$

$$\frac{a_m^j}{s_m^i} \cdot \left| \frac{a_m^j}{a_v^j + a_m^j + a_s^j} - \frac{s_m^i}{s_v^i + s_m^i + s_s^i} \right| +$$

$$\frac{a_s^j}{s_s^i} \cdot \left| \frac{a_s^j}{a_v^j + a_m^j + a_s^j} - \frac{s_s^i}{s_v^i + s_m^i + s_s^i} \right|$$

To account for the possibility that jobs with similar comparative profiles are characterized by different overall skill intensities and individuals with similar comparative profiles are characterized by different overall ability levels, we also create an absolute profile:

$$AA_{i,j} = \frac{a_v^j + a_m^j + a_s^j}{s_v^i + s_m^i + s_s^i}$$

Given the non-linearity of payoffs in our model, the distinction between comparative and absolute advantage is critical for our calibration strategy.

Finally, combining absolute and comparative dimensions, we construct a job-specific ability:

$$AA_{i,j} - CA_{i,j}.$$

5 Quantitative Analysis

5.1 Calibration Strategy

In the calibration exercise we parametrize the economy in order to match aggregate and micro moments that characterize the US economy in 1970, 1990, and 2010 to their model counterparts. We select these years since the respondents from Project TALENT, NLSY79, and NLSY97 are, on average, 29 years old at these three dates. The parameterization treats these three target years as steady states of the model. We restrict our sample to two groups: men and women. We exclude individuals who report to be unemployed or on active military duty.

The Project TALENT 11-year follow up reports information on 29-year-old individuals who attended at least one year of high school. We select equivalent samples from the NLSY79 and NLSY97. The “matching” Census and ACS samples include individuals who attended at least one year of high school and are in the 25-34 age group.

Respondents are assigned to home production if they are out of the labor force or if they work less than 15 hours per week. Individuals working at least 30 hours are classified as employed in one of 20 occupations. Those who work between 15 and 29 hours are classified as part-time workers. We split their sampling weight equally between home production and the reported market occupations.

We measure earnings as the sum of labor, business, and farm income in the previous year. To calculate the earnings, we restrict the sample to those employed at least 48 weeks per year, working at least 30 hours per week, and earning at least 1,000 dollars in 2010 dollars in the previous year. We convert all earnings data from the Census to constant dollars.

We assume that the idiosyncratic occupation-specific abilities are drawn from a Fréchet distribution. The draws are i.i.d. across occupations and the distribution is identical across groups. Since workers currently participate in the labor market for a single period, we cannot disentangle cohort from time effects and this, in turn, implies that we cannot distinguish of educational barriers from labor market discrimination. For this reason we assume that one is facing any educational barriers. In addition, we assume that men face no labor market discrimination, while female workers face labor market barriers in all occupations except teaching. We normalize the elasticities of human capital with respect to ability (α) and class size $\left(\frac{\beta}{\sigma}\right)$ to one. Table 4 below summarizes our assumptions and reports all normalized parameters.

To estimate θ and η to infer non-teaching occupational productivities.

Given our assumption that idiosyncratic abilities are i.i.d. Fréchet draws it follows that wages within the same occupation and group follow a Fréchet distribution with shape parameter $\theta(1-\eta)$. Using Census and ACS data, we estimate the best fit of $\theta(1-\eta)$ using a cross-sectional

Parameter	Definition	Determination	Value
$\tau_{o,men}^w$	labor market barriers for men	assumption	0
$\tau_{o,g}^e$	human capital barriers for all groups	assumption	0
$\tau_{T,g}^w$	labor market barriers in teaching (all groups)	assumption	0
$\tau_{T,g}^e$	human capital barriers in teaching (all groups)	assumption	0
α	elasticity of human capital with respect to idiosyncratic ability	normalization	1
A_{20}	Productivity in home production	normalization	1
σ	elasticity of human capital with respect to class size	free parameter	0.5
β	elasticity of human capital with respect to teacher's human capital	free parameter	0.5

Table 4: Assumptions and Normalized Parameters

regression of log hourly residual wages on $20 \times 2 \times 10$ occupation-group-age dummies in each year. The estimate for $\theta(1 - \eta)$ is 1.324, which is consistent with values found in the earlier literature. The parameter η denotes the elasticity of human capital with respect to human capital investment (in terms of goods) and is equal to the fraction of output spent on human capital accumulation. Public plus private spending on education as a share of U.S. GDP is, on average, 6.6 percent in 1995, 2000, and 2010. Since the labor share in the United States in the same three years was 0.641, this implies an η of 0.103. Given our estimate for $\theta(1 - \eta)$ this implies $\theta=1.476$.

Our assumption on the distribution of idiosyncratic abilities, the human capital accumulation technology, and the functional form of the teachers' wage profile imply that the wages of teachers (within a group) follow a Fréchet distribution with shape parameter $\theta(1 - \gamma\eta)$. we estimate the best fit of $\theta(1 - \gamma\eta)$ using a cross-sectional regression of log hourly residual wages on 2×10 group-age dummies in each year. The estimated value of $\theta(1 - \gamma\eta)$ is 1.350. Given the estimates for θ and η , this implies $\gamma=0.830$.

We pick ϕ to match the Mincerian return from a cross-sectional regression of log income on years of schooling (with occupation $\times$ group dummies). Our estimate for ϕ is 2.745. The average schooling of teachers (in terms of educational attainment) relative to the average schooling of other workers is 1.23. This implies $\mu = 0.714$.

A summary of all time-invariant parameters is in Table 5 below.

Parameter	Definition	Determination	Value
θ	shape parameter of Fréchet-distributed idiosyncratic abilities	wage dispersion in non-teaching occupations (indirect inference)	1.476
η	goods elasticity of human capital	aggregate education spending share (indirect inference)	0.103
ϕ	time elasticity of human capital	Mincerian returns to education (non-teaching) (indirect inference)	2.745
γ	curvature of wage function in teaching	wage dispersion in teaching (indirect inference)	0.83
μ	trade-off between consumption and time spent accumulating human capital	schooling of teachers relative to schooling of others	0.714

Table 5: Baseline Parameter

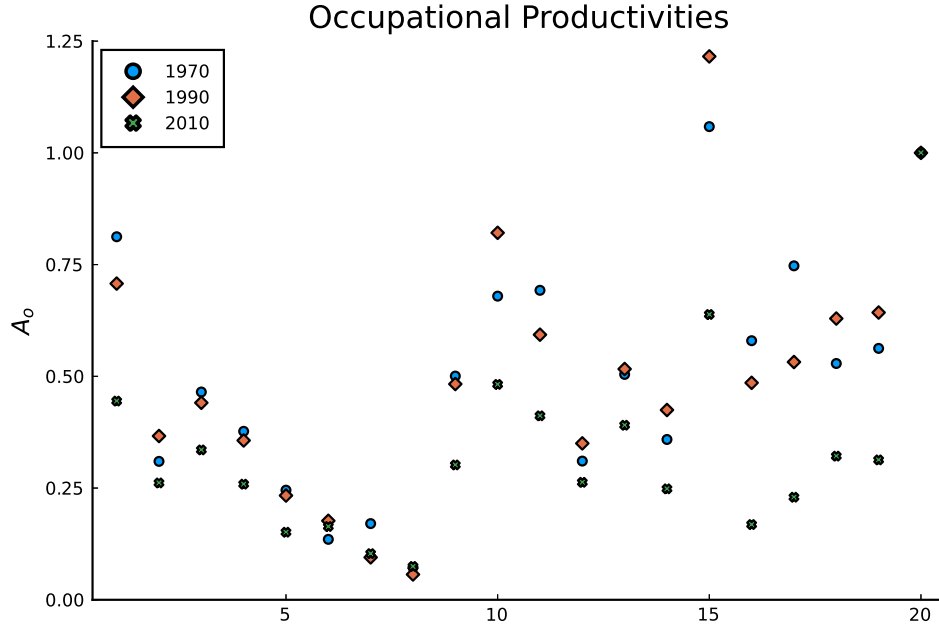


Figure 7: Occupational Productivities over Time

Occupational productivities are recovered from men's occupational shares in 1970, 1990, and 2010. Without loss of generality, we normalize the productivity of home production (occupation 20) to one each year. Figure 7 illustrates occupational productivities over time. The downward trend in productivities are a reflection of the gradual rise in the share of home production (which includes workers who are not in the labor force) among men.

We identify the extent of labor market discrimination against women in non-teaching occupations based on differences in the occupational shares of women relative to men year-by-year and we plot them in Figure 8. In general, the values of τ_w for women drift towards zero over time, albeit it with some exceptions. The “subsidy” on home production (occupation 20) falls sharply from 1970 to 2010, which reflects the sizeable reduction in the proportion of women engaged in home production over this time.

The scale parameter of the wage function in teaching, i.e., κ , determines the fraction of men who select into teaching. The calibrated values are 0.289, 0.190, and 0.169 for 1970, 1990, and 2010, respectively. This downward trend reflects the gradual drop in this share from 2.4 percent in 1970 to 1.9 in 2010.

The parameter that governs the extent of aggregate labor market discrimination against women in non-teaching occupations, λ_f , is imputed from the fraction of females who select teaching as their occupation. The calibrated steady-state values are 0.879, 1.034, and 0.602 for 1970,

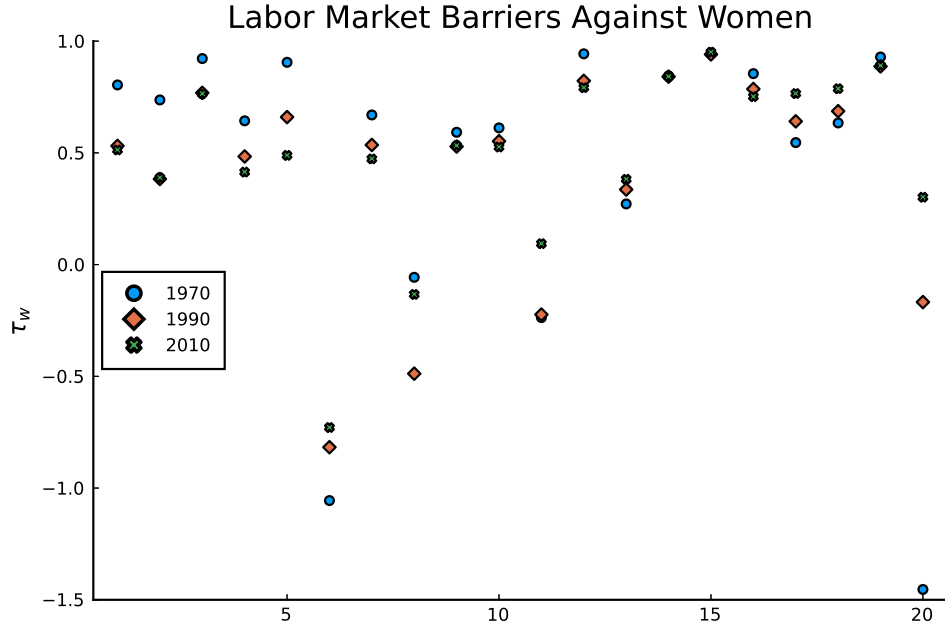


Figure 8: Labor Market Barriers over Time

1990, and 2010, respectively.

Table 6 summarizes the parameters that vary over time.

Parameter	Definition	Determination
A_o	occupational productivities (non-teaching)	labor market shares for men
$\tau_{o,women}^w$	labor market discrimination (non-teaching) against women	labor market shares for women
κ	scale parameter of wage function in teaching	fraction of males who are teachers
λ_f	aggregate labor market discrimination against women in non-teaching occupations	fraction of females who are teachers

Table 6: Parameter Varying over Time

5.2 Calibration Results

The evolution of occupational productivities and labor market discrimination – captured by both τ^w and λ_f – gives rise to significant changes in the economy’s human capital intensity and skill composition of teachers. While the skill composition of female teachers is broadly stable between 1970 and 1990, it shifts to the left between 1990 and 2010 (Figure 9). The skill distribution of male teachers, on the other hand, shifts to the right at first and then partially

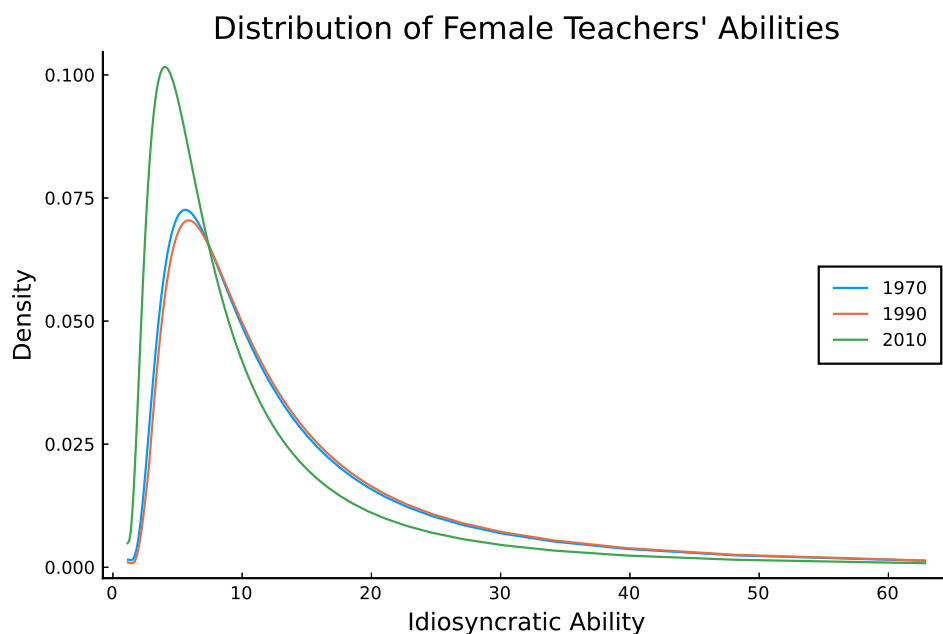


Figure 9: Female Teachers

reverts to the left between 1990 and 2010 (Figure 9).

Since women face no labor market discrimination in teaching, human capital investment by male and female teachers is identical once we control for ability (see Figure 11). Strikingly, however, human capital investment falls sharply between 1970 and 1990. It stabilizes at this low level between 1990 and 2010. Together, the evolution of the skill distribution among teachers and of the human capital investment imply that the drop in human capital *per teacher* declines by 15 percent between 1970 and 1990. It drops an additional 9 percent between 1990 and 2010.

The drop in human capital per teacher is offset by a rise in the number of teachers such that $\tilde{H}_T$ is about the same in 1970 and 2010. Equation 9 implies that class sizes are shrinking over time, which is indeed what we observe in the data.

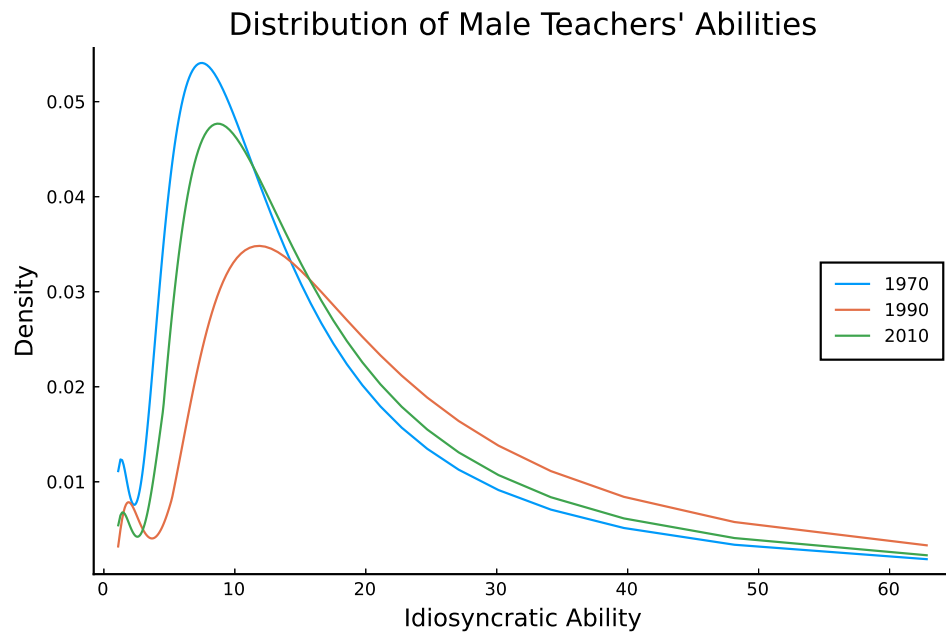


Figure 10: Male Teachers

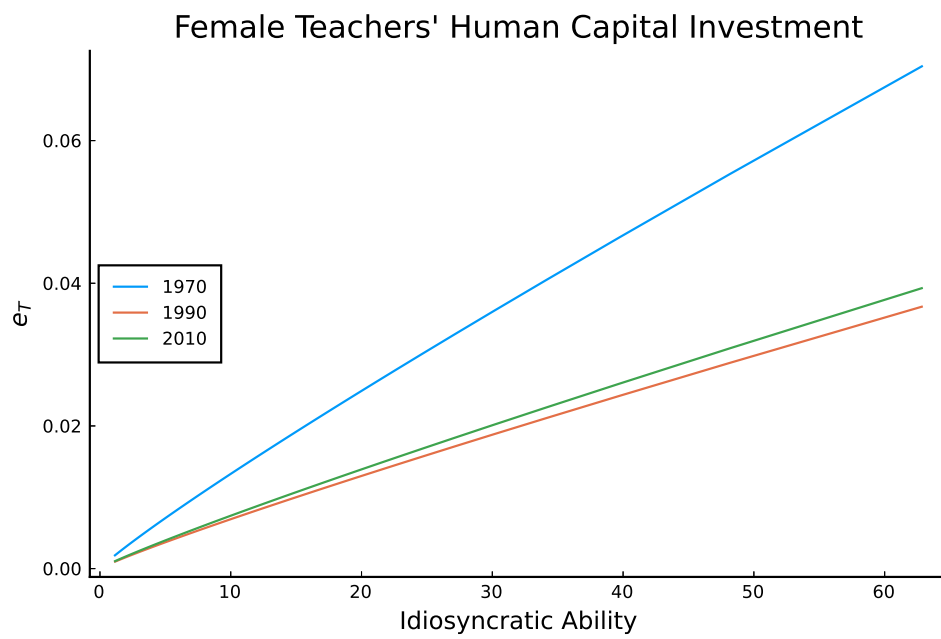


Figure 11: Human Capital Investment (Men and Women)

6 Conclusion

This paper develops a novel occupational choice model that captures the distinct nature of teaching and learning – and hence human capital accumulation. In addition to the static gains and losses associated with labor market distortions and educational barriers, we can characterize the dynamic effects that are driven by the number and ability of workers who are drawn to the teaching profession. This occupational choice, in turn, shapes the incentives to invest in human capital when the agents are young, i.e. when they are students.

The model also introduces wage profiles that are non-linear in the workers’ human capital and this generalization implies that occupational choice is driven by *comparative* as well as *absolute* advantages across the various occupations. This is a significant departure from the canonical occupational choice models in the spirit of ?. At this point, the extent of this non-linearity is not pinned down precisely and this is an area of ongoing work.

The model reconciles several salient stylized facts. In addition to the change in the student-teacher ratios and per-student-expenditures mentioned above, they are: (1) the increase in the share of women in teaching jobs from 4.6 percent in 1970 to 6.7 percent in 2010; (2) the drop in the share of men who are teachers from 2.9 percent to 2.1 percent during that same time; (3) the sharp rise in the female labor force participation rate; (4) the slight decline in the male labor force participation rate; and (5) the evolution of the skill composition by gender and occupation between 1970 and 2010.

The somewhat loose discussion of our calibration strategy in section 5.1 reflects the work-in-progress nature of our quantitative exercise. The model is otherwise well identified and in ongoing work we are calibrating the remaining parameters to match the salient moments at three different points in time: 1970, 1990, and 2010. We select these years since the respondents from all three longitudinal surveys are, on average, 29 years old. This parameterization will treat these three target years as steady states of the model. Future work will explore the human capital transition dynamics between steady states.

In ongoing work, we are introducing multiple locations to address the stylized fact that there arguably is some sorting between the quality of teachers and the abilities of students across (rather than within) school or school districts. ? analyses the effect of school finance equalization on student outcomes and intergenerational mobility. In the context of our model, the introduction of multiple locations relaxes the rigidity of teachers’ wage contracts: high-income households with, on average, higher-ability children will agglomerate and broaden the tax base the funds teachers’ salaries locally and high-ability teachers can move to wealthy districts rather than quit teaching altogether. Positive sorting between teachers and students (households) affects aggregate outcomes through two channels with countervailing effects:

on the one hand, competition between districts can attract high-quality workers into teaching, on the other hand, the segregation between low and high-ability students increases inequality. Our framework will be able to decompose the effects cleanly.

A List of Occupations

1. Executives, Administrative, and Managerial
2. Management Related
3. Architects, Engineers, Math, and Computer Science
4. Natural and Social Scientists, Recreation, Religious, Arts, Athletes
5. Doctors and Lawyers
6. Nurses, Therapists, and Other Health Service
7. Teachers, Postsecondary
8. Teachers, Non-Postsecondary and Librarians
9. Health and Science Technicians
10. Sales, All
11. Administrative Support, Clerks, Record
12. Fire, Police, and Guards
13. Food, Cleaning, and Personal Services and Private Household
14. Farm, Related Agriculture, Logging, and Extraction
15. Mechanics and Construction
16. Precision Manufacturing
17. Manufacturing Operators
18. Fabricators, Inspectors, and Material Handlers
19. Vehicle Operators
20. Home Production
21. Kindergarten - Secondary Teachers

B Special Case: Teacher's Are Paid Their Marginal Product

The model in section 2 assumes that teachers are paid according to an exogenous schedule ω_T and each teacher's level of pay is a function of her human capital h_T . Here we consider a special case where teachers are paid according to their contribution to the human capital acquisition of the students in their classroom. Since the assignment of students to teachers is random and equally-abled students acquire the same amount of human capital across classrooms "run" by different teachers, better teachers are compensated for the additional human capital they contribute to by virtue of differences in class size.

In a nutshell, this implies that teachers are paid according to the wage profile in Proposition 4.

B.1 Human Capital Investment

Let's first characterize the optimal human capital investment for a prospective teacher with ability a from group g .

The F.O.C.s for $s_{T,g}$ and $e_{T,g}$, respectively, after a few steps of algebra are:

$$(1 + \tau_{T,g}^e) e_{T,g} \left(\frac{\sigma}{\eta\beta} - 1 \right) = (1 - s_{T,g}) \mu (1 - t') (1 - \tau_{T,g}^{\omega'}) \frac{\partial \omega'}{\partial s_{T,g}} \quad (31)$$

$$(1 - t') (1 - \tau_{T,g}^{\omega'}) \frac{\partial \omega'}{\partial e_{T,g}} = 1 + \tau_{T,g}^e, \quad (32)$$

where

$$\frac{\partial \omega'}{\partial s_{T,g}} = \frac{\phi\beta}{\sigma} \lambda' \left(\frac{M}{2\tilde{H}_T} \right)^{-\beta} \frac{M}{2\tilde{H}_T'} \left[a^\alpha (s_{T,g})^\phi (e_{T,g})^\eta \right]^{\frac{\beta}{\sigma}} \frac{1}{s_{T,g}} \quad (33)$$

$$\frac{\partial \omega'}{\partial e_{T,g}} = \frac{\eta\beta}{\sigma} \lambda' \left(\frac{M}{2\tilde{H}_T} \right)^{-\beta} \frac{M}{2\tilde{H}_T'} \left[a^\alpha (s_{T,g})^\phi (e_{T,g})^\eta \right]^{\frac{\beta}{\sigma}} \frac{1}{e_{T,g}} \quad (34)$$

Let $\tau_{T,g} \equiv \frac{(1-t')(1-\tau_{T,g}^{\omega'})}{1+\tau_{T,g}^e}$ be a composite wedge that takes into account the rate at which income is taxed, the extent of labor market discrimination, and educational barriers. Note that $\tau_{T,g} = 1$ when $t' = \tau_{T,g}^{\omega'} = \tau_{T,g}^e = 0$.

The amount of time and goods this individual invests in human capital is then given by:

$$s_{T,g} = \frac{\mu\phi}{\mu\phi + \frac{\sigma}{\beta} - \eta} \quad (35)$$

$$e_{T,g} = \frac{\beta}{\sigma} \cdot \eta^{\frac{1}{1-\eta}} \cdot \left(\frac{2\tilde{H}_T}{M}\right)^{\frac{\sigma}{1-\eta}} \cdot \frac{\sum_{g=1}^G A'_O{}^{\frac{1}{1-\eta}} \cdot \tau_{O,g}^{\frac{\eta}{1-\eta}} \cdot s_{O,g}^{\frac{\phi}{1-\eta}} \cdot \int_0^\infty a^{\frac{\alpha}{1-\eta}} f_{O,g}(a) da}{\sum_{g=1}^G f_{O,g}(a) da} \\ \times \frac{\tau_{T,g}^{\frac{\sigma}{\sigma-\eta\beta}} \cdot s_{T,g}^{\frac{\phi\beta}{\sigma-\eta\beta}} \cdot a_T^{\frac{\alpha\beta}{\sigma-\eta\beta}}}{\sum_{g=1}^G \tau_{T,g}^{\frac{\eta\beta}{\sigma-\eta\beta}} \cdot s_{T,g}^{\frac{\phi\beta}{\sigma-\eta\beta}} \cdot \int_0^\infty a^{\frac{\alpha\beta}{\sigma-\eta\beta}} f_{g,T}(a) da} \quad (36)$$

Note that a prospective teacher's optimal investment of resources depends on the total human capital investment of prospective *production* workers. Clearly, it depends explicitly on $s_{O,g}$, but it also depends on $e_{O,g}$ in equation (38) below, albeit less obviously so. This captures the fact that teachers are compensated for their contribution to the accumulation of human capital that can be used for the production of the consumable final good.¹⁹

The F.O.C.s for prospective production workers (i.e., those labeled “other”) are analogues of (35) and (36) with the simplification that the wage profile in production (as opposed to teaching) is linear in $h_{O,g}$ and the optimal amount of time and resources invested in human capital is given by:

$$s_{O,g} = \frac{\mu\phi}{\mu\phi + 1 - \eta} \quad (37)$$

$$e_{O,g} = \left(\tau_{O,g} \cdot \eta \cdot \left(\frac{2\tilde{H}_T}{M}\right)^\sigma \cdot A'_O{}^\eta \cdot s_{O,g}^\phi \cdot a_O^\alpha \right)^{\frac{1}{1-\eta}} \quad (38)$$

where, analogously to the problem of a prospective teacher, we define $\tau_{O,g} \equiv \frac{(1-t')(1-\tau_{O,g}^{\omega'})}{1+\tau_{O,g}^e}$.

In the special case with $\beta = \sigma$, the model reverts to a standard occupational choice model à la Roy and the wage profile is linear in human capital on *both* occupations.

Proposition 3 *If $\beta = \sigma$, then $\frac{d\omega}{dh}$ and $\frac{h^*(h_O)}{h_O}$ are constant, i.e. the occupational cutoff satisfies $h^* =$*

¹⁹When the economy has $I > 2$ occupations, the prospective teacher's optimal investment of resources is characterized by:

$$e_{T,g} = \frac{\beta}{\sigma} \cdot \eta^{\frac{1}{1-\eta}} \cdot \left(\frac{2\tilde{H}_T}{M}\right)^{\frac{\sigma}{1-\eta}} \cdot \frac{\sum_{i=2}^I \sum_{g=1}^G A'_i{}^{\frac{1}{1-\eta}} \cdot \tau_{i,g}^{\frac{\eta}{1-\eta}} \cdot s_{i,g}^{\frac{\phi}{1-\eta}} \cdot \int_0^\infty a^{\frac{\alpha}{1-\eta}} f_{i,g}(a) da}{\sum_{i=2}^I \sum_{g=1}^G f_{i,g}(a) da} \\ \times \frac{\tau_{T,g}^{\frac{\sigma}{\sigma-\eta\beta}} \cdot s_{T,g}^{\frac{\phi\beta}{\sigma-\eta\beta}} \cdot a_T^{\frac{\alpha\beta}{\sigma-\eta\beta}}}{\sum_{g=1}^G \tau_{T,g}^{\frac{\eta\beta}{\sigma-\eta\beta}} \cdot s_{T,g}^{\frac{\phi\beta}{\sigma-\eta\beta}} \cdot \int_0^\infty a^{\frac{\alpha\beta}{\sigma-\eta\beta}} f_{g,T}(a) da}.$$

$\frac{h_T}{h_O}$ for all h_O .

B.2 The Aggregate Laws of Motion

The economy is characterized by the following laws of motions: ²⁰

$$\begin{aligned} \tilde{H}'_T = & \left[\left(\frac{\beta}{\sigma} \right)^\eta \cdot \eta^{\frac{1}{1-\eta}} \cdot \left(\frac{2\tilde{H}_T}{M} \right)^{\frac{\sigma}{1-\eta}} \cdot \left(\frac{\sum_{i=2}^I \sum_{g=1}^G A'_i{}^{\frac{1}{1-\eta}} \cdot \tau_{i,g}^{\frac{\eta}{1-\eta}} \cdot s_{i,g}^{\frac{\phi}{1-\eta}} \cdot \int_0^\infty a^{\frac{\alpha}{1-\eta}} f_{i,g}(a) da}{\sum_{i=2}^I \sum_{g=1}^G f_{i,g}(a) da} \right)^\eta \right. \\ & \left. \times \left(\sum_{g=1}^G \tau_{T,g}^{\frac{\eta\beta}{\sigma-\eta\beta}} \cdot s_{T,g}^{\frac{\phi\beta}{\sigma-\eta\beta}} \cdot \int_0^\infty a^{\frac{\alpha\beta}{\sigma-\eta\beta}} f^{g,T}(a) da \right)^{\frac{\sigma-\eta\beta}{\beta}} \right]^{\frac{\beta}{\sigma}} \end{aligned} \quad (39)$$

$$H'_O = \eta^{\frac{\eta}{1-\eta}} \cdot \left(\frac{2\tilde{H}_T}{M} \right)^{\frac{\sigma}{1-\eta}} \cdot \sum_{g=1}^G \tau_{O,g}^{\frac{\eta}{1-\eta}} \cdot \int_0^\infty A'_O{}^{\frac{\eta}{1-\eta}} \cdot s_{O,g}^{\frac{\phi}{1-\eta}} \cdot a^{\frac{\alpha}{1-\eta}} f_{O,g}(a) da. \quad (40)$$

B.3 Occupational Wages in Decentralized Competitive Equilibrium

In order to highlight the mechanisms that govern the individuals' occupational choice, we first solve for a decentralized equilibrium that implements the optimal allocation of adults to "teaching" and "other". Put differently, we solve the model for a wage profile $\omega^*(h_{T,g})$ that pays teachers with human capital $h_{T,g}$ their marginal product.²¹

How can we characterize the teacher's marginal product? Recall that only workers produce the final (consumption) good using a linear production function with productivity A_O . Thanks to stable random matching of students to teachers, the ability distribution of students is identical across class rooms and a teacher's marginal value only varies with the number of students in her class room, denoted by $N(h_{T,g})$. Clearly, then, the wage profile is linear in $N(\cdot)$. Since N is not linear in $h_{T,g}$ in general this implies that the wage profile is not linear in $h_{T,g}$ either.

²⁰When the economy has $I > 2$ occupations, the law of motion for the aggregate human capital in teaching is characterized by:

$$\begin{aligned} \tilde{H}'_T = & \left[\left(\frac{\beta}{\sigma} \right)^\eta \cdot \eta^{\frac{1}{1-\eta}} \cdot \left(\frac{2\tilde{H}_T}{M} \right)^{\frac{\sigma}{1-\eta}} \cdot \left(\frac{\sum_{g=1}^G A'_O{}^{\frac{1}{1-\eta}} \cdot \tau_{O,g}^{\frac{\eta}{1-\eta}} \cdot s_{O,g}^{\frac{\phi}{1-\eta}} \cdot \int_0^\infty a^{\frac{\alpha}{1-\eta}} f_{O,g}(a) da}{\sum_{g=1}^G f_{O,g}(a) da} \right)^\eta \right. \\ & \left. \times \left(\sum_{g=1}^G \tau_{T,g}^{\frac{\eta\beta}{\sigma-\eta\beta}} \cdot s_{T,g}^{\frac{\phi\beta}{\sigma-\eta\beta}} \cdot \int_0^\infty a^{\frac{\alpha\beta}{\sigma-\eta\beta}} f^{g,T}(a) da \right)^{\frac{\sigma-\eta\beta}{\beta}} \right]^{\frac{\beta}{\sigma}} \end{aligned}$$

²¹In section 2 we specified the model for an arbitrary profile ω . Here we impose further restrictions on this function.

Since teachers are compensated for the incremental human capital in “other”, we can characterize the teacher’s wage profile.

Proposition 4 (Teacher’s Wage Profile) *For a given teacher with human capital $h_{T,g}$ and aggregate human capital in teaching $\tilde{H}_T$, the wage in teaching is proportional to the number of students in the teacher’s class and is characterized by:*

$$\omega(h_{T,g}, \tilde{H}_T) = \frac{\overbrace{H'_O A'_O}^{\text{total output (next period)}}}{\underbrace{\int_0^\infty f_{O,g}(a) da}_{\text{fraction of prospective production workers in class}}} \cdot \underbrace{\frac{(h_{T,g})^{\frac{\beta}{\sigma}}}{\tilde{H}_T}}_{\text{fraction of students taught}} \quad (41)$$

where

$$\tilde{H}_T = \sum_{g=1}^G \int_0^\infty h_{T,g}^{\frac{\beta}{\sigma}} f_{T,g}(a) da$$

is the $\left(\frac{\beta}{\sigma}\right)^{th}$ moment of the distribution of human capital among current teachers.

The second term in equation (41) captures the idea that a teacher’s wage depends on class size, which is a non-linear function of her human capital $h_{T,g}$. Recall from equations (33) and (34) that a prospective teacher’s optimal $s_{T,g}$ and $e_{T,g}$ depends on the quality of today’s teachers (i.e. their human capital) and on the slope of the wage profile next period and hence on both today’s $\tilde{H}_T$ as well as tomorrow’s $\tilde{H}'_T$.²²

$$\begin{aligned} \omega = & \eta^{\frac{\eta}{1-\eta}} \cdot \left(\frac{2\tilde{H}_T}{M}\right)^{\frac{\sigma}{1-\eta}} \cdot \frac{\sum_{g=1}^G A'_O{}^{\frac{1}{1-\eta}} \cdot \tau_{O,g}^{\frac{\eta}{1-\eta}} \cdot s_{O,g}^{\frac{\phi}{1-\eta}} \cdot \int_0^\infty a^{\frac{\alpha}{1-\eta}} f_{O,g}(a) da}{\sum_{g=1}^G \int_0^\infty f_{O,g}(a) da} \\ & \times \frac{\tau_{T,g}^{\frac{\eta\beta}{\sigma-\eta\beta}} \cdot s_{T,g}^{\frac{\phi\beta}{\sigma-\eta\beta}} \cdot a_T^{\frac{\alpha\beta}{\sigma-\eta\beta}}}{\sum_{g=1}^G \tau_{T,g}^{\frac{\eta\beta}{\sigma-\eta\beta}} \cdot s_{T,g}^{\frac{\phi\beta}{\sigma-\eta\beta}} \cdot \int_0^\infty a^{\frac{\alpha\beta}{\sigma-\eta\beta}} f_{g,T}(a) da} \end{aligned} \quad (42)$$

²²When the economy has $I > 2$ occupations, the teachers’ wage profile is characterized by:

$$\begin{aligned} \omega = & \eta^{\frac{\eta}{1-\eta}} \cdot \left(\frac{2\tilde{H}_T}{M}\right)^{\frac{\sigma}{1-\eta}} \cdot \frac{\sum_{i=2}^I \sum_{g=1}^G A'_i{}^{\frac{1}{1-\eta}} \cdot \tau_{i,g}^{\frac{\eta}{1-\eta}} \cdot s_{i,g}^{\frac{\phi}{1-\eta}} \cdot \int_0^\infty a^{\frac{\alpha}{1-\eta}} f_{i,g}(a) da}{\sum_{i=2}^I \sum_{g=1}^G \int_0^\infty f_{i,g}(a) da} \\ & \times \frac{\tau_{T,g}^{\frac{\eta\beta}{\sigma-\eta\beta}} \cdot s_{T,g}^{\frac{\phi\beta}{\sigma-\eta\beta}} \cdot a_T^{\frac{\alpha\beta}{\sigma-\eta\beta}}}{\sum_{g=1}^G \tau_{T,g}^{\frac{\eta\beta}{\sigma-\eta\beta}} \cdot s_{T,g}^{\frac{\phi\beta}{\sigma-\eta\beta}} \cdot \int_0^\infty a^{\frac{\alpha\beta}{\sigma-\eta\beta}} f_{g,T}(a) da} \end{aligned}$$

B.4 Occupational Choice

To highlight the effect of educational barriers and labor market discrimination we add group- and occupation-specific distortions to the model. This is mostly a heuristic exercise since parsimony of our state space enables us to characterize the full transition path from one set of distortions to another. We highlight this feature of the model here since there have arguably been secular changes in educational barriers and labor market distortions for groups other than white males. To fix ideas, however, we simply compare steady states associated with particular wedges τ indexed by ω (for labor market discrimination), e (for educational barriers), and g for groups (race, gender, ...).

We can show that this economy has a steady state in H_T and we can characterize it in a single equation:²³

$$\begin{aligned} \tilde{H}_T = & \left[\left(\frac{\beta}{\sigma} \right)^\eta \cdot \eta^{\frac{1}{1-\eta}} \cdot \left(\frac{2}{M} \right)^{\frac{\sigma}{1-\eta}} \cdot \left(\frac{\sum_{g=1}^G A_i'^{\frac{1}{1-\eta}} \cdot \tau_{O,g}^{\frac{\eta}{1-\eta}} \cdot s_{O,g}^{\frac{\phi}{1-\eta}} \cdot \int_0^\infty a^{\frac{\alpha}{1-\eta}} f_{O,g}(a) da}{\sum_{g=1}^G f_{O,g}(a) da} \right)^\eta \right. \\ & \left. \times \left(\sum_{g=1}^G \tau_{T,g}^{\frac{\eta\beta}{\sigma-\eta\beta}} \cdot s_{T,g}^{\frac{\phi\beta}{\sigma-\eta\beta}} \cdot \int_0^\infty a^{\frac{\alpha\beta}{\sigma-\eta\beta}} f^{g,T}(a) da \right)^{\frac{\sigma-\eta\beta}{\beta}} \right]^{\frac{\beta}{\sigma} \cdot \frac{1-\eta}{1-\eta-\beta}} \end{aligned} \quad (43)$$

Proposition 5 (Occupational Choice Boundary) *The occupational choice boundary, denoted $\bar{a}_{T,g}(a_O)$, depends on the magnitude of the educational barriers and labor market frictions, but does not depend*

²³When the economy has $I > 2$ occupations, the steady state H_T is characterized by:

$$\begin{aligned} \tilde{H}_T = & \left[\left(\frac{\beta}{\sigma} \right)^\eta \cdot \eta^{\frac{1}{1-\eta}} \cdot \left(\frac{2}{M} \right)^{\frac{\sigma}{1-\eta}} \cdot \left(\frac{\sum_{i=2}^I \sum_{g=1}^G A_i'^{\frac{1}{1-\eta}} \cdot \tau_{i,g}^{\frac{\eta}{1-\eta}} \cdot s_{i,g}^{\frac{\phi}{1-\eta}} \cdot \int_0^\infty a^{\frac{\alpha}{1-\eta}} f_{i,g}(a) da}{\sum_{i=2}^I \sum_{g=1}^G f_{i,g}(a) da} \right)^\eta \right. \\ & \left. \times \left(\sum_{g=1}^G \tau_{T,g}^{\frac{\eta\beta}{\sigma-\eta\beta}} \cdot s_{T,g}^{\frac{\phi\beta}{\sigma-\eta\beta}} \cdot \int_0^\infty a^{\frac{\alpha\beta}{\sigma-\eta\beta}} f^{g,T}(a) da \right)^{\frac{\sigma-\eta\beta}{\beta}} \right]^{\frac{\beta}{\sigma} \cdot \frac{1-\eta}{1-\eta-\beta}} \end{aligned}$$

on either current or future aggregate human capital in teaching $\widetilde{H}_T$. It is characterized by:²⁴

$$\begin{aligned}
& \frac{a_O^{\frac{\alpha}{1-\eta}}}{a_T^{\frac{\alpha\beta}{\sigma-\eta\beta}}} \cdot \frac{\tau_{O,g}^{\frac{1}{1-\eta}}}{\tau_{T,g}^{\frac{\sigma}{\sigma-\eta\beta}}} \cdot \frac{1-\eta}{1-\frac{\beta\eta}{\sigma}} \cdot \frac{1+\tau_{O,g}^e}{1+\tau_{T,g}^e} \cdot \left(\frac{1-s_{O,g}}{1-s_{T,g}} \right)^{\frac{1}{\mu}} \cdot \frac{s_{O,g}^{\frac{\phi}{1-\eta}}}{s_{T,g}^{\frac{\phi\beta}{\sigma-\eta\beta}}} \\
&= \frac{\sum_{g=1}^G A'_O{}^{\frac{1}{1-\eta}} \cdot \tau_{O,g}^{\frac{\eta}{1-\eta}} \cdot s_{O,g}^{\frac{\phi}{1-\eta}} \cdot \int_0^\infty a^{\frac{\alpha}{1-\eta}} f_{O,g}(a) da}{A'_O{}^{\frac{1}{1-\eta}} \cdot \sum_{g=1}^G \tau_{T,g}^{\frac{\eta\beta}{\sigma-\eta\beta}} \cdot s_{T,g}^{\frac{\phi\beta}{\sigma-\eta\beta}} \cdot \int_0^\infty a^{\frac{\alpha\beta}{\sigma-\eta\beta}} f_{T,g}(a) da \cdot \sum_{g=1}^G \int_0^\infty f_{O,g}(a) da} \quad (44)
\end{aligned}$$

²⁴When the economy has $I > 2$ occupations, the occupational choice boundary is characterized by:

$$\begin{aligned}
& \frac{a_i^{\frac{\alpha}{1-\eta}}}{a_T^{\frac{\alpha\beta}{\sigma-\eta\beta}}} \cdot \frac{\tau_{i,g}^{\frac{1}{1-\eta}}}{\tau_{T,g}^{\frac{\sigma}{\sigma-\eta\beta}}} \cdot \frac{1-\eta}{1-\frac{\beta\eta}{\sigma}} \cdot \frac{1+\tau_{i,g}^e}{1+\tau_{T,g}^e} \cdot \left(\frac{1-s_{i,g}}{1-s_{T,g}} \right)^{\frac{1}{\mu}} \cdot \frac{s_{i,g}^{\frac{\phi}{1-\eta}}}{s_{T,g}^{\frac{\phi\beta}{\sigma-\eta\beta}}} \\
&= \frac{\sum_{i=2}^I \sum_{g=1}^G A'_i{}^{\frac{1}{1-\eta}} \cdot \tau_{i,g}^{\frac{\eta}{1-\eta}} \cdot s_{i,g}^{\frac{\phi}{1-\eta}} \cdot \int_0^\infty a^{\frac{\alpha}{1-\eta}} f_{i,g}(a) da}{A'_i{}^{\frac{1}{1-\eta}} \cdot \sum_{g=1}^G \tau_{T,g}^{\frac{\eta\beta}{\sigma-\eta\beta}} \cdot s_{T,g}^{\frac{\phi\beta}{\sigma-\eta\beta}} \cdot \int_0^\infty a^{\frac{\alpha\beta}{\sigma-\eta\beta}} f_{T,g}(a) da \cdot \sum_{i=2}^I \sum_{g=1}^G \int_0^\infty f_{i,g}(a) da}
\end{aligned}$$